



Republic of the Philippines
PROVINCIAL GOVERNOR'S OFFICE
Communication & Information Services Office
Pagadian City, Zamboanga del Sur
-oOo-



PROJECT PROPOSAL

On

Victory FREE WiFi **Internet sa Barangay Project**

Initiated by:

PROVINCIAL GOVERNMENT
OF ZAMBOANGA DEL SUR

Submitted by:

JUSTIN AHMED PAOLO C. HERRERA
Head, ICT Operations
PGO-CISO



Table of Contents

Cover Page	1
Executive Summary	2
Project Description	4
Project Workplan	5
Human Resources	6
Project Sites	8
Project Monitoring	9
Proposed Budget	10
Signatories	16
Attachment 1 – Pagadian City Sites	17
Attachment 2 – Labangan Sites	18
Attachment 3 – Tukuran Sites	19
Attachment 4 – Aurora Sites	20
Attachment 5 – R. Magsaysay Sites	21
Attachment 6 – Tambulig Sites	22
Attachment 7 – Molave Sites	23
Attachment 8 – Mahayag Sites	24
Attachment 9 – Josefina Sites	25
Attachment 10 – Dumingag Sites	26
Attachment 11 – Sominot Sites	27
Attachment 12 – Midsalip Sites	28
Attachment 13 – Dumalinao Sites	29
Attachment 14 – Guipos Sites	30
Attachment 15 – San Pablo Sites	31
Attachment 16 – Dinas Sites	32
Attachment 17 – Dimataling Sites	33
Attachment 18 – Tabina Sites	34
Attachment 19 – Pitogo Sites	35
Attachment 20 – V. Sagun Sites	36
Attachment 21 – Margosatubig Sites	37
Attachment 22 – Lapuyan Sites	38
Attachment 23 – San Miguel Sites	39
Attachment 24 – Kumalarang Sites	40
Attachment 25 – Bayog Sites	41
Attachment 26 – Lakewood Sites	42
Attachment 27 – Tigbao Sites	43
Attachment 28 – Backbone Link	44
Attachment 29 – Captive Portal	45

Executive Summary

In conjunction with the COVID-19 pandemic, internet access is a basic need nowadays to support government programs, especially in the education sector. The DepEd will implement distant learning and online modules to promote social distancing.

Part of the essential services is internet access. But it seems, most of the remote communities are lacking about it.

The national government, through DICT, however, has ongoing Public WiFi Projects, but only 102 sites were initiated out of 681 barangays for the entire province.

Because of this, the Provincial Governor made an initiative to break these barriers by proposing a Province-Wide Internet Project that can be available for every barangay.

ORGANIZATION

This project was organized mainly for public welfare, through the effort of the energetic leadership of Engr. Victor J. Yu, the provincial governor of Zamboanga del Sur with his spirited behalf Divina Grace Yu, Congresswoman of 1st District and the technical team of the Communication and Information Services Office, and with the technical support from the Department of Information and Communication Technology (DICT) made this happen in reality.

PROJECT DESCRIPTION

This project shall be called Victory FREE WiFi. The scope of this project is province-wide and shall cater 600+ barangays of about 100,000 users more or less. The general public (users) has access, and through it can connect to other networks or the Internet.

The network datacenter, management, and control are at the provincial capitol building. (Network Topologies as described in the attachments).

The infrastructure uses mostly wireless technology utilizing the 2.4Ghz and 5Ghz band from the bridge, base stations, and the access points.

PROJECT WORKPLAN

Implementation duration shall be in 60 calendar days upon approval. Activities are the following:

- Acquire Leased Line Agreement from ISP on the negotiated bandwidth.
- Site Surveys and Inventory of existing tower (*Previously Barangay Telephone System*)
- Procuring Materials and Equipment.
- The hiring of Workforce (Human Resource).
- Training / Workshop on Network Installation.
- Configuration - Equipment IP Pairing and Testing.
- Onsite Installation - Router Bridges, Base Stations, CPEs, Access Points, and Extenders.
- Dry Run - Network Management, Testing, and Fine Tuning.
- Launching - Project Launching, MOA Signing, and Turn-Over to Stakeholders.
- Creation of Project Maintenance Team.

HUMAN RESOURCES

1. Technical Working Group

- 1 Chief of Operation - Provides Logistics Support and Over-All Project Manager.
- 1 Programmer for Network Management - Monitors Manage Network Links and Status.
- 1 Team Leader for Net Plans & Deployment - Conduct Site Surveys and Supervise Installations.
- 1 IT Assistant for Net Plans & Deployment - Site Surveys, Assist Network Device Installers.
- 1 IT Assistant for Network Deployment - Site Surveys, Assist Network Device Installers.
- 1 Logistic Driver - Onsite Coordinator, Assist On-Site Technicians.

2. Device Installers

- 20 Riggers - Device and Antenna Installer (with knowledge in the field).
- 10 IT Associates - Manage Installation, Termination, and Cabling.
- 10 Workers - Equipment Keeper, Loader, and Lifter.
- Dispatch Driver - Equipment and Personnel Transporter

3. Project Monitoring

- Project Maintenance Team
- Clustered Barangay Facilitator (CBF) serves as an integral part of a project maintenance team.

PROJECT SITES

Project Sites are located mostly in the province of Zamboanga del Sur. Still, there is a site that traversed to the other area, specifically in the municipality of Buug to complete the Network Topology System.

Sites include the Data Center located at the Provincial Capitol Building, then to Central Base Station at Mt. Palpalan, then to 27 municipalities with two (2) or more Municipal Base Stations and 600 more or less Barangay Access Points and 300+ AP Extenders. *(Please refer to the attached Project Sites for Geographical location).*

PROJECT MONITORING

Network Infrastructure shall be monitored remotely through the Ubiquiti Network Management System (UNMS) by any or an authorized member of the Project Maintenance Team with CBF functioning as an onsite correspondent for efficient system repair and maintenance.

PROPOSED BUDGET

• Network Devices	-	Php	35,826,498.70
• Installation of Equipment (2 Months)	-	Php	255,348.50
• Internet Bandwidth (2 Years Subscription)	-	Php	2,400,000.00
• Site Survey Inventory & Planning (1 Month)	-	Php	90,000.00
• Human Resources (2 Months)	-	Php	1,440,000.00
• Monitoring & Maintenance (2 Years)	-	Php	3,360,000.00
• Overhead Cost (Electrical Bills & Other Devices)	-	Php	5,000,000.00
TOTAL	-	Php	48,319,337.20

I. PROJECT DESCRIPTION

A. Background

Amidst the COVID19 pandemic, the provincial government of Zamboanga del has never hampered by its commitment to deliver essential services to the constituents.

Part of the essential services is internet access. But it seems, most of the remote communities are lacking about it.

The national government, through DICT, however, has ongoing Public WiFi Projects, but only 102 sites were initiated out of 600+ barangays for the entire province.

In this regard, the provincial government of Zamboanga del Sur took a collaborative effort with the concerned technical agency and other resources to break these barriers.

This proposed project dubbed as Province-Wide Internet that can be made available for every barangays. Concerning Republic Act 10929, otherwise known as the "Free Internet Access in Public Places Act" signed by President Rodrigo Roa Duterte.

B. Purpose and Scope

In conjunction with the COVID-19 pandemic, internet access is a basic need nowadays to support government programs, especially in the education sector. The DepEd soon will implement distant learning and online modules to promote social distancing.

This project shall be called Victory FREE WiFi and shall be the solution to the nowadays problem. The scope of this project is province-wide and shall cater 600+ barangays of 27 municipalities or about 100,000 users more or less. The general public (users) has access, and through it can connect to other networks or the Internet.

C. Technical

The network infrastructure uses mostly wireless technology utilizing the 2.4Ghz and 5Ghz band from the bridge, base stations, and to the access points and extenders.

The network datacenter, management, and control are at the provincial capitol building. Network Topologies were described in the attachments.

The public (Users) with a captive portal (authentication) can access the WiFi Hotspot access points wirelessly within a range of 500 meters radius with the managed speed of up to 2mbps approximately and can browse essential web content. Clean browsing or content filtering shall be imposed to avoid exploitation of the network bandwidth.

Cache Server, Bandwidth Management, User Management, and other network solutions shall be utilized to maximize system performance.

II. PROJECT WORKPLAN

Project Workplan duration shall be within 60 days upon approval of the project. Below is the work plan to wit:

- A. Acquire/Install Leased Line Agreement from ISP on the negotiated bandwidth.
- B. Site Surveys and Inventory of existing tower (*Previously Barangay Telephone System*)
- C. Procuring Materials and Equipment.
- D. The hiring of Workforce (Human Resource).
- E. Training / Workshop on Network Installation.
- F. Configuration – Equipment IP Pairing and Testing.
- G. Onsite Installation – Router Bridges, Base Stations, CPEs, Access Points, and Extenders.
- H. Dry Run – Network Management, Testing, and Fine Tuning.
- I. Launching – Project Launching, MOA Signing, and Turn-Over to Stakeholders.
- J. Creation of Project Maintenance Team.

PROJECT ACTIVITY MATRIX

ID	Activity	Specific Task	Responsible	Days
A	Bandwidth Acquisition	Negotiate Leased Line (Dedicated) Bandwidth	ISP (PLDT) PGO-CISO	2
B	Site Survey	Site Survey and Inventory of Existing Towers	PGO-CISO-DICT	60
C	Procurement	Procurement of Equipment, Materials, and Tools	PGO-CISO	15
D	Human Resource	Hiring of Riggers, IT Associates, Utility Workers	PGO-CISO	5
E	Training/Workshop	Network Installation Training of Human Resources	PGO-CISO-TWG	5
F	Configuration	Equipment Configuration and Testing (IP Pairing)	PGO-CISO-TWG	5
G	Installation	Servers, Switch, Routers, AP, Antenna Installations	CISO Human Resources	30
H	Dry Run	Network Management, Testing & Configurations	PGO-CISO-TWG	5
I	Project Launching	Project Launching, MOA Signing, and Turn-Over	PGO-CISO-DICT-ISP	1
J	Project Maintenance	Creation of Project Maintenance & Monitoring	PGO-CISO-TWG	1

PROJECT TIME FRAME

ID	Activity/Task	Start	Finish	%
A	Bandwidth Acquisition – Negotiate Leased Line (Dedicated) Bandwidth	06/10/2020	06/20/2020	100
B	Site Survey – Site Survey and Inventory of Existing Towers	06/01/2020	07/30/2020	75
C	Procurement – Procurement of Equipment, Materials, and Tools	07/15/2020	07/30/2020	50
D	Human Resource – Hiring of Riggers, IT Associates, Utility Workers	07/15/2020	07/20/2020	-
E	Training/Workshop – Network Installation Training of Human Resources	07/20/2020	06/30/2020	-
F	Configuration – Equipment Configuration and Testing (IP Pairing)	08/01/2020	08/05/2020	-
G	Installation – Servers, Switch, Routers, AP, Antenna Installations	08/01/2020	08/30/2020	-
H	Dry Run – Network Management, Testing & Configurations	08/25/2020	08/30/2020	-
I	Project Launching – Project Launching, MOA Signing, and Turn-Over	08/25/2020	08/30/2020	-
J	Project Maintenance – Creation of Project Maintenance & Monitoring	Continuing	Continuing	-

III. HUMAN RESOURCES

A. Technical Working Group

- 1 Chief of Operation – Provides Logistics Support and Over-All Project Manager.
- 1 Programmer for Network Management – Monitors Manage Network Links and Status.
- 1 Team Leader, Plans & Deployment – Conduct Site Surveys and Supervise Installations.
- 1 IT Assistant for Net Plans & Deployment – Site Surveys, Assist Network Device Installers.
- 1 IT Assistant for Network Deployment – Site Surveys, Assist Network Device Installers.
- 1 Logistic Driver – Onsite Coordinator, Assist On-Site Technicians.

#	Name of TWG Member	Nature of Work	Responsibility
1	Justine Paolo Herrera	Chief of Operation	Provides Logistics Support and Over-All Project Manager
2	Arex Pantallano	Network Administrator	Monitors and Manage the Network Links and Status
3	Jaime Castillon Maata	On-Site Supervisor	Supervise Network Installation, Repair, and Maintenance
4	Rolly Joy F. Duhaylungsod	IT, Onsite Technician for Plans and Deployment	Assist Plan Preparation, Onsite Installation and Surveys
5	Junrey Saman	IT, Onsite Technician for Plans and Deployment	Assist Plan Preparation, Onsite Installation and Surveys
6	Troy Perez	Logistics and Transportation	Assist Onsite Technicians, provide transportation, Etc.

B. Device Installers

- 20 Riggers – Device and Antenna Installer (with knowledge in the field).
- 10 Associates in IT – Manage Installation, Termination, and Cabling.
- 10 Utility Workers – Equipment Keeper, Loader, and Lifter.
- 1 Dispatch Driver – Equipment and Personnel transporter.

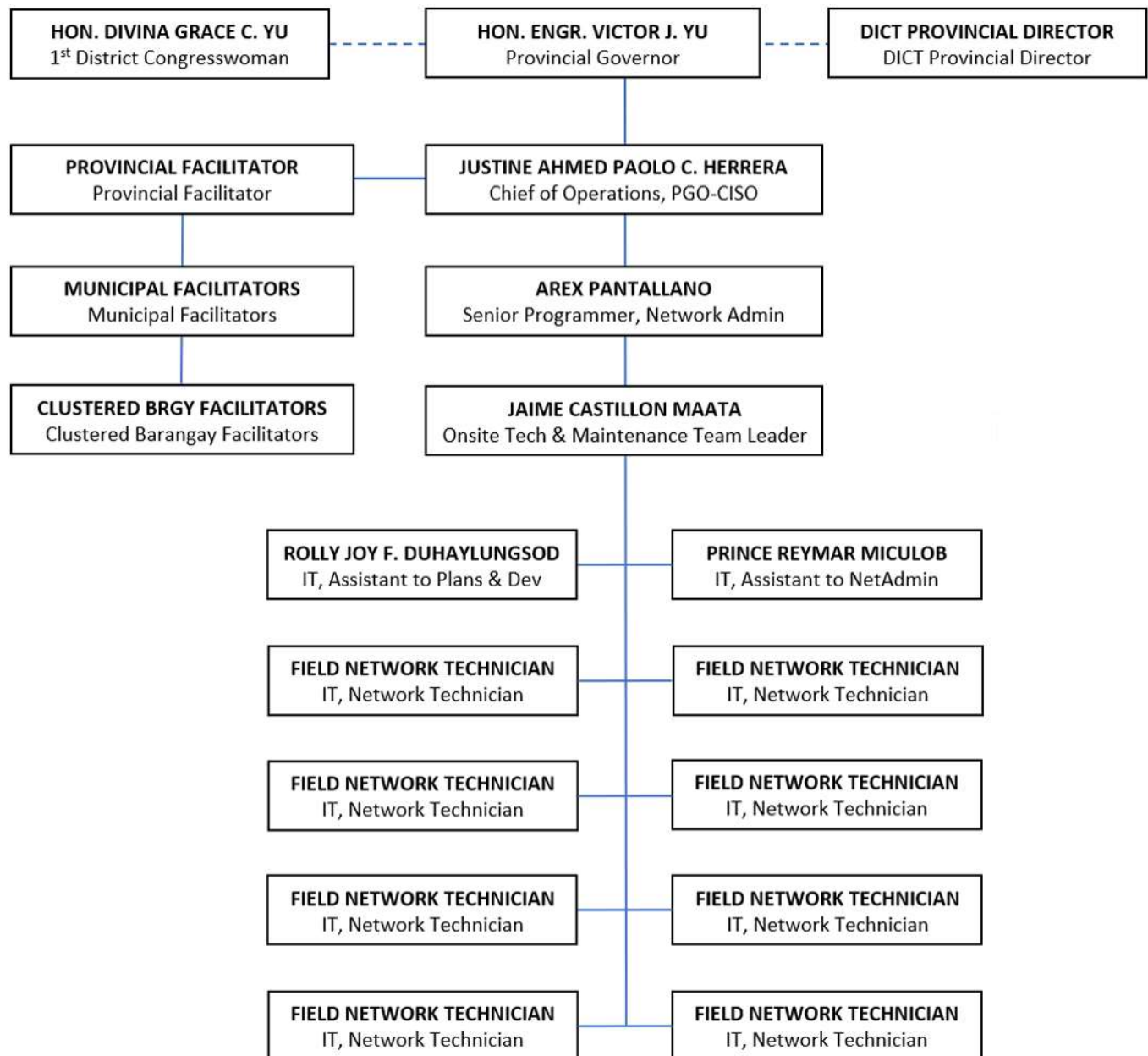
C. Maintenance

- Project Maintenance Team – Ensure the functionality of the network and fix system failures.
- Clustered Barangay Facilitator (CBF) serves as an integral part of a project maintenance team.

D. Organizational Structure

This project was organized mainly for public welfare, through the effort of the energetic leadership of Engr. Victor J. Yu, the provincial governor of Zamboanga del Sur with his strong-willed behalf Divina Grace Yu, Congresswoman of 1st District and the technical team of the Communication and Information Services Office, and with the technical support from the Department of Information and Communication Technology (DICT) made this happen in reality.

ORGANIZATIONAL STRUCTURE



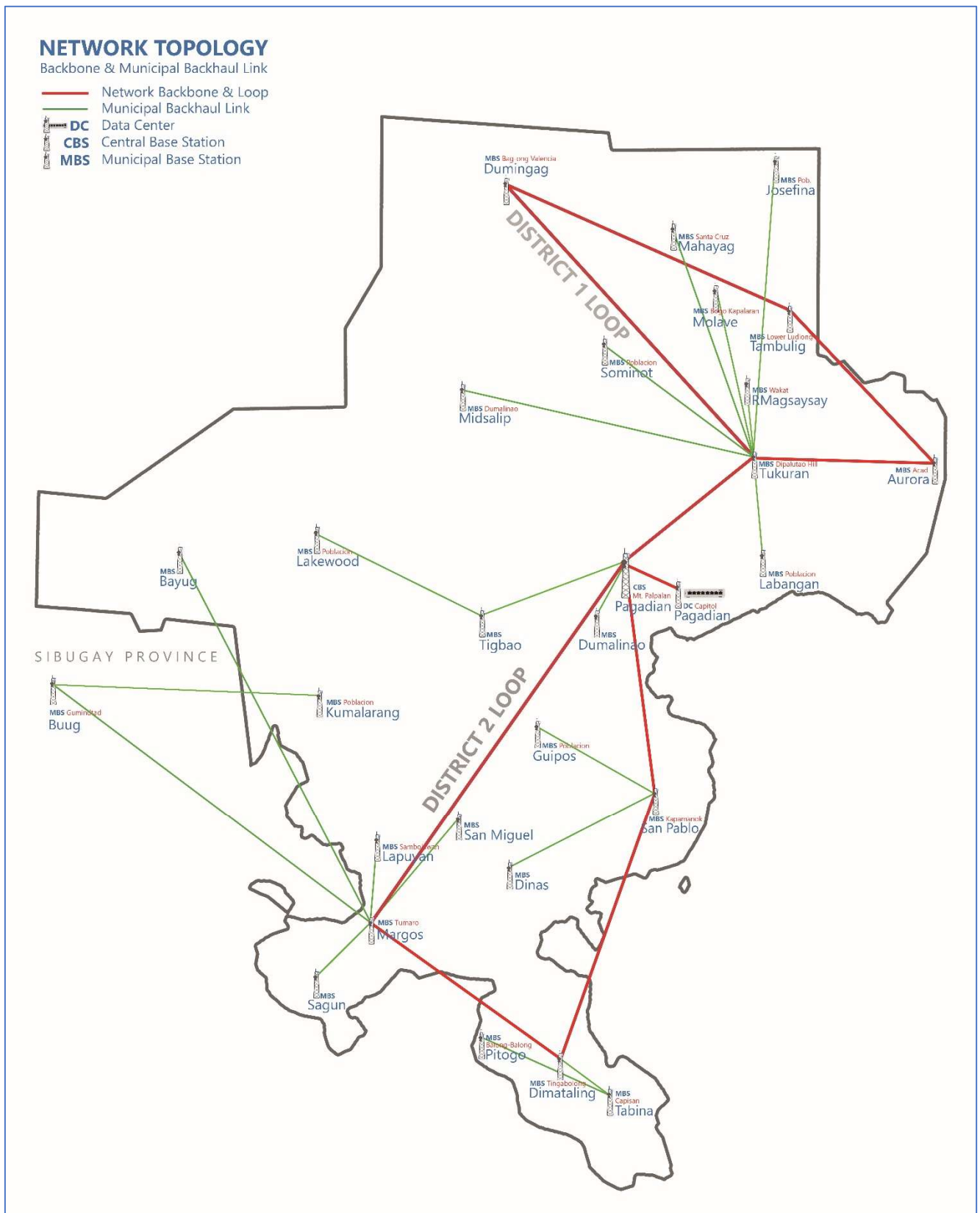
IV. PROJECT SITES

Project Sites are located mostly in the province of Zamboanga del Sur. Still, there is a site that traversed to the other area, specifically in the municipality of Buug to complete the Network Topology System.

Sites include the Data Center located at the Provincial Capitol Building, then to Central Base Station at Mt. Palpalan, then to 27 municipalities with two (2) or more Municipal Base Stations and 600 more or less Barangay Access Points and 300+ AP Extenders. *(Please refer to the attached Project Sites for Geographical location).*

NETWORK TOPOLOGY

Illustration below shows the network topology Backbone, how every municipality is being connected to the network, and its geographical locations. (*Barangay Access Points are drawn in separate sheets*).



V. PROJECT MONITORING

Network Infrastructure shall be monitored remotely through the Ubiquiti Network Management System (UNMS) by any or an authorized member of the Project Maintenance Team with MFs and CBFs functioning as an onsite correspondent.

Fig. 3. UNMS Interface



Using UNMS, a network administrator at a single location can manage groups of Ubiquiti devices: monitor the network, track usage, and make changes to more than one device at a time. UNMS offers these features:

- airOS® Device Monitoring and Management
- Mass Configuration for airOS 5 and airOS 6
- Mass Firmware Upgrade
- Network Topology Visualization
- Setup of Alerts and Notifications
- Smart Grouping and Filtering of Devices (using Device Groups)
- Device Discovery and Automatic Network Topology Resolution
- Configurable Charts
- Tasks and Task Scheduling
- Configurable Device List (Control View)
- Map Support
- Ping, Speed Test
- Malware Cleanup Utility

UNMS supports almost all devices used by the project like the Ubiquiti® airMAX® M series, airMAX AC series, and airFiber® series products of which Victory FREE WiFi project uses most of these products.

VI. PROPOSED BUDGET

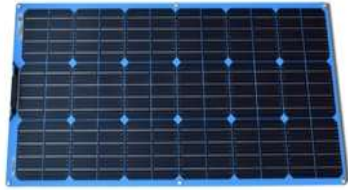
Total project cost is Forty Eight Million Three Hundred Nineten Thousand Three Hundred Thirty-Seven Pesos and 20th (Php **48,319,337.20**), including planning, site surveys, human resources, and Equipment. Devices are based on Manufacturer's Suggested Retail Price (MSRP) quotation as of **June 28, 2020**. Currency rate at USD 1 = PHP 53.00.

A. NETWORK DEVICE

#	Description & Reference	Image	QTY	SRP(USD)
1	airFiber 5 GHz High-Band Bridge https://store.ui.com/collections/operator-airfiber/products/airfiber5u		14	\$ 999.99
2	airMAX Rocket AC 5 GHz Radio Gen 2 https://store.ui.com/collections/wireless/products/rocket-5ac-prism-gen2?_pos=1&_sid=16cdf1f29&_ss=r		200	\$ 249.00
3	airMAX AC Sector 5 GHz, 45°, 22 dBi Antenna https://store.ui.com/collections/operator-airmax-and-ltu/products/5ghz-airmax-basestation-22dbi-45-deg-ac		100	\$ 199.00
4	airMAX® RD-5G34 2x2 PtP Bridge Dish Antenna https://store.ui.com/collections/wireless/products/5ghz-rocketdish-34dbi-rocket-kit?_pos=4&_sid=c8b1afc1b&_ss=r		100	\$ 329.00

5	airMAX AMO-2G13 Omni 2.4GHz 13dBi Antenna https://store.ui.com/collections/wireless/products/2-4ghz-airmax-omni-13dbi-rocket-kit?_pos=1&_sid=04994aa9c&_ss=r		300	\$ 245.00
6	airMAX PowerBeam M5 GHz, 400 mm Bridge https://store.ui.com/collections/wireless/products/powerbeam-m5-400?_pos=4&_sid=dee409f43&_ss=r		100	\$ 95.00
7	airMAX LiteBeam AC 5 GHz Bridge SKU: LBE-5AC-GEN2-US https://store.ui.com/collections/wireless/products/litebeam-m5?_pos=5&_sid=28484e422&_ss=r		600	\$ 49.00
8	UniFi® AC Mesh Access Point https://store.ui.com/collections/wireless/products/unifi-ac-mesh-ap?_pos=2&_sid=643caca00&_ss=r		1,000	\$ 99.00
9	Cloud Core Router CCR1036-12G- 4S-EM https://mikrotik.com/product/CCR1036-12G-4S-EM		1	\$ 1,195.00
10	Edge Router Infinity 16-Core, 1.8 GHz Processor https://store.ui.com/collections/routing-switching/products/edgerouter-8-xg?_pos=1&_sid=1a560128c&_ss=r		1	\$ 1,849.00
11	Edge Switch 16 XG, 320 Gbps Switching Capacity https://store.ui.com/collections/routing-switching/products/edgeswitch-16-xg?_pos=1&_sid=6b666cec2&_ss=r		13	\$ 669.00

12	UniFi Switch, 8-Port, Non-POE https://store.ui.com/collections/unifi-network-routing-switching/products/unifi-switch-8		30	\$ 99.00
13	Outdoor 4-Port PoE Passthrough Nano Switch https://store.ui.com/collections/routing-switching/products/nanoswitch?_pos=1&sid=8320003be&ss=r		600	\$ 40.00
14	UF-RJ45-10G Ethernet to SFP Ports Module https://store.ui.com/collections/related/products/uf-rj45-10g-1?_pos=1&sid=434594a68&ss=r		30	\$ 65.00
15	UniFi Ethernet Patch Cable UC-Patch-RJ45-BL https://store.ui.com/collections/unifi-accessories-cables/products/unifi-ethernet-patch-cable-with-bendable-booted-rj45?_pos=1&sid=f3fd96553&ss=r		100	\$ 2.00
16	Network TOUGHCable Connectors (Pack of 100) https://www.bhphotovideo.com/c/product/1049759-REG/ubiquiti_networks_tc_con_100_tough_cable_connectors_pack_of.html		30	\$ 52.00
17	Carrier Grade TOUGHCable 1000 Ft https://store.ui.com/collections/operator-accessories/products/toughcable-carrier		5	\$ 180.00
18	HW6845BIG Outdoor Network Rack Cabinet https://harwell.en.alibaba.com/product/60756681241-801825058/Outdoor_Electrical_Power_Distribution_Rack_Cabinets_For_Pole_Mount.html		30	\$ 156.00

19	200W 12V Solar Panel with 12V 10A Solar Charge Controller https://www.aliexpress.com/item/4000197247114.html		60	\$ 207.00
20	B.B Rechargeable VRLA Lead Acid Battery 12v 40ah https://www.lazada.com.ph/products/brandnew-original-bb-rechargeable-vrla-lead-acid-battery-12v-40ah-for-solar-ups-back-up-power-electric-vehicles-i579042148-s1580328759.html?spm=a2o4l.searchlist.list.5.744e2c6eyrJmG1&search=1		60	\$ 62.00
21	3000W DC 12V to AC 220V Solar Voltage Inverter https://www.lazada.com.ph/products/solar-inverter-3000w-peak-voltage-transformer-converter-dc-12v-to-ac-220v-car-inverter-for-solar-inverter-home-appliances-i758828250-s2272676284.html?spm=a2o4l.searchlist.list.79.20a8387d5PCAmP&search=1		60	US\$ 26.00
22	Desktop Computer for Virtualization/IP Management/DHCP Server, Cache Server and Device Controller Specs: 2pcs 2TB SSD, 32GB RAM, Intel Core i7 or Zeon 8th Gen CPU		1	US\$ 660.37
23	Laptop Computer for Network Configuration/Management (For Installation) Specs: At least Intel Core i7 8th Gen CPU, 16MB RAM, 1TB SSD.		2	US\$ 754.72

Network Device (Recap)

NO.	QTY.	DESCRIPTION	MSRP	TOTAL
1	14	airFiber® 5 AF5, 5GHz, 1Gbps+, FDD, 100Km+ Radio	90,009.90	1,260,138.60
2	200	airMAX® Rocket AC 5 GHz Radio Gen 2	22,434.90	4,486,980.00
3	100	airMAX® AC Sector 5 GHz, 45°, 22 dBi Antenna	17,929.90	1,792,990.00
4	100	airMAX® RD-5G34 2x2 PtP Bridge Dish Antenna	29,642.90	2,964,290.00
5	300	airMAX® AMO-2G13 Omni 2.4GHz 13dBi Antenna	22,074.50	6,622,350.00
6	100	airMAX® PowerBeam M5 GHz, 400 mm Bridge	8,559.50	855,950.00
7	600	airMAX® LiteBeam AC 5 GHz Bridge	4,414.90	2,648,940.00
8	1000	UniFi® AC Mesh Access Point	8,919.90	8,919,900.00
9	1	Cloud Core Router CCR1036-12G-4S-EM	107,669.50	107,669.50
10	1	Edge Router Infinity 16-Core, 1.8 GHz Processor	166,594.90	166,594.90
11	13	EdgeSwitch 16 XG 320 Gbps Switching Capacity	56,876.90	739,399.70
12	30	UniFi® Switch, 8-Port, Non-POE	8,919.90	267,597.00
13	600	Outdoor 4-Port PoE Passthrough Nano Switch	3,604.00	2,162,400.00
14	30	UF-RJ45-10G Ethernet to SFP Ports Module	5,856.50	175,695.00
15	100	UniFi® Ethernet Patch Cable UC-Patch-RJ45-BL	180.20	18,020.00
16	30	Network TOUGH Cable Connectors (Pack of 100)	4,685.20	140,556.00
17	5	Carrier Grade TOUGH Cable 1000 Ft	16,218.00	81,090.00
18	30	HW6845BIG Outdoor Network Rack Cabinet	14,055.60	421,668.00
19	60	200W 12V Solar Panel with 12V 10A Solar Charge Controller	18,650.70	1,119,042.00
20	60	Rechargeable VRLA Lead Acid Battery 12v 40ah	5,586.20	335,172.00
21	60	3000W DC 12V To AC 220V Solar Voltage Inverter	2,342.60	140,556.00
22	1	Set Desktop Computer, 32GB RAM, 2TB SSD, 19" Monitor	127,500.00	127,500.00
23	2	Laptop Computer, 16GB RAM, 1TB SSD, 8 th Gen 17" Display	136,000.00	272,000.00
SUB-TOTAL (A)			PHP	35,826,498.70

B. INSTALLATION EQUIPMENT

NO.	QTY.	DESCRIPTION	MSRP	TOTAL
1	20	Universal Cable Crimping Tools	495.55	9,911.00
2	20	Nylon CABLE TIE (1,000 Pcs)	427.97	8,559.50
3	10	10-120X80 Long Range Zoom Hunting Telescope	4,505.00	45,050.00
4	20	Full Body Safety Belt Harness with Lanyard Hook	1,711.90	34,238.00
5	40	Universal Wrench	180.20	7,208.00
6	40	Adjustable Wrench	270.30	10,812.00
7	500	BS1139 Pressed Scaffolding Swivel Clamp/Coupler	62.90	31,450.00
8	100	Scaffolding G.I. Pipe Schedule 20 1 1/2 x 6 meters	1,081.20	108,120.00
SUB-TOTAL (B)				255,348.50

C. INTERNET BANDWIDTH (24 Months Lease)

NO.	MOS	DESCRIPTION	PRICE	TOTAL
1	24	300 Mbps Leased (Dedicated) Line Burstable to 600 Mbps	100,000.00	2,400,00.00
SUB-TOTAL (C)				2,400,000.00

D. SITE SURVEY, INVENTORY & PLANNING

NO.	DAYS	DESCRIPTION	PER DIEM	TOTAL
1	30	Occular Inspection and Inventory of Existing Towers	1,000.00	30,000.00
2	30	Drafting of Plans and Project Proposal	1,000.00	30,000.00
3	30	Allowances / Fuel	1,000.00	30,000.00
SUB-TOTAL (D)				90,000.00

E. HUMAN RESOURCES (Installation)

NO.	DAYS	DESCRIPTION	WAGES	TOTAL
1	60	Wages of 20 Device Installer/Rigger @ P500.00 per Day	20,000.00	600,000.00
2	60	Wages of 10 IT Technicians @ P500.00 per Day	10,000.00	300,000.00
3	60	Wages of 10 Utility Workers @ P400.00 per Day	8,000.00	240,000.00
4	60	Miscellaneous and Other Expenses	10,000.00	300,000.00
SUB-TOTAL (E)				1,440,000.00

F. MONITORING & MAINTENANCE

NO.	MOS	DESCRIPTION	SALARY	TOTAL
1	24	Salary of 1 Onsite Network Administrator	20,000.00	480,000.00
2	24	Salary of 8 Onsite Network Technician	15,000.00	2,880,000.00
SUB-TOTAL (F)				3,360,000.00

G. OVERHEAD COST (Electrical Bills, Network Rack, UPS and Other Expenses)

NO.	LOT	DESCRIPTION	ESTIMATE	TOTAL
1	1	Electrical Bills, Network Rack, UPS and Other Expenses	5,000,00.00	5,000,000.00
SUB-TOTAL (G)				5,000,000.00

RECAPITULATIONS:

A. Network Devices	-	Php	35,826,498.70
B. Installation Equipment	-	Php	255,348.50
C. Internet Bandwidth (2 Years Subscription)	-	Php	2,400,000.00
D. Site Survey Inventory & Planning (1 Month)	-	Php	90,000.00
E. Human Resources (2 Months)	-	Php	1,440,000.00
F. Monitoring & Maintenance (2 Years)	-	Php	3,360,000.00
G. Overhead Cost (Electrical Bills & Other Expenses)	-	Php	5,000,000.00
GRAND TOTAL	-	Php	48,319,337.20

Prepared by:

THE TECHNICAL WORKING GROUP

Victory FREE WiFi Project
Network Plans and Development

Reviewed by:

JUSTINE AHMED PAOLO C. HERRERA

Head, ICT Operations
PGO-CISO

AREX PANTALLANO

Network Administrator
Senior Programmer, PGO-CISO

Recommending Approval:

THE DICT DIRECTOR

DICT Provincial Director
Zamboanga del Sur

HON. DIVINA. GRACE C. YU

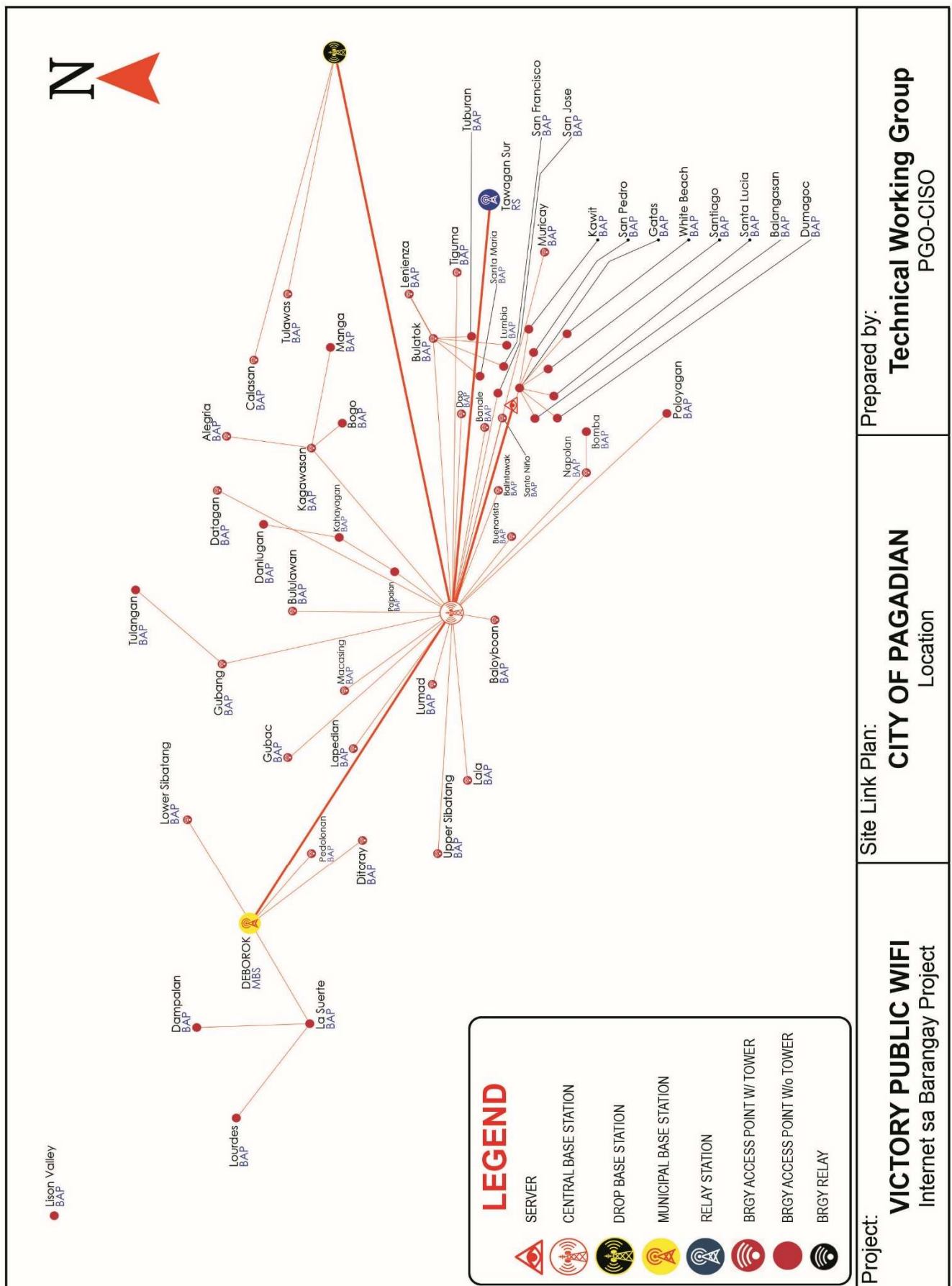
Congresswoman, 1st District
Zamboanga del Sur

APPROVED BY:

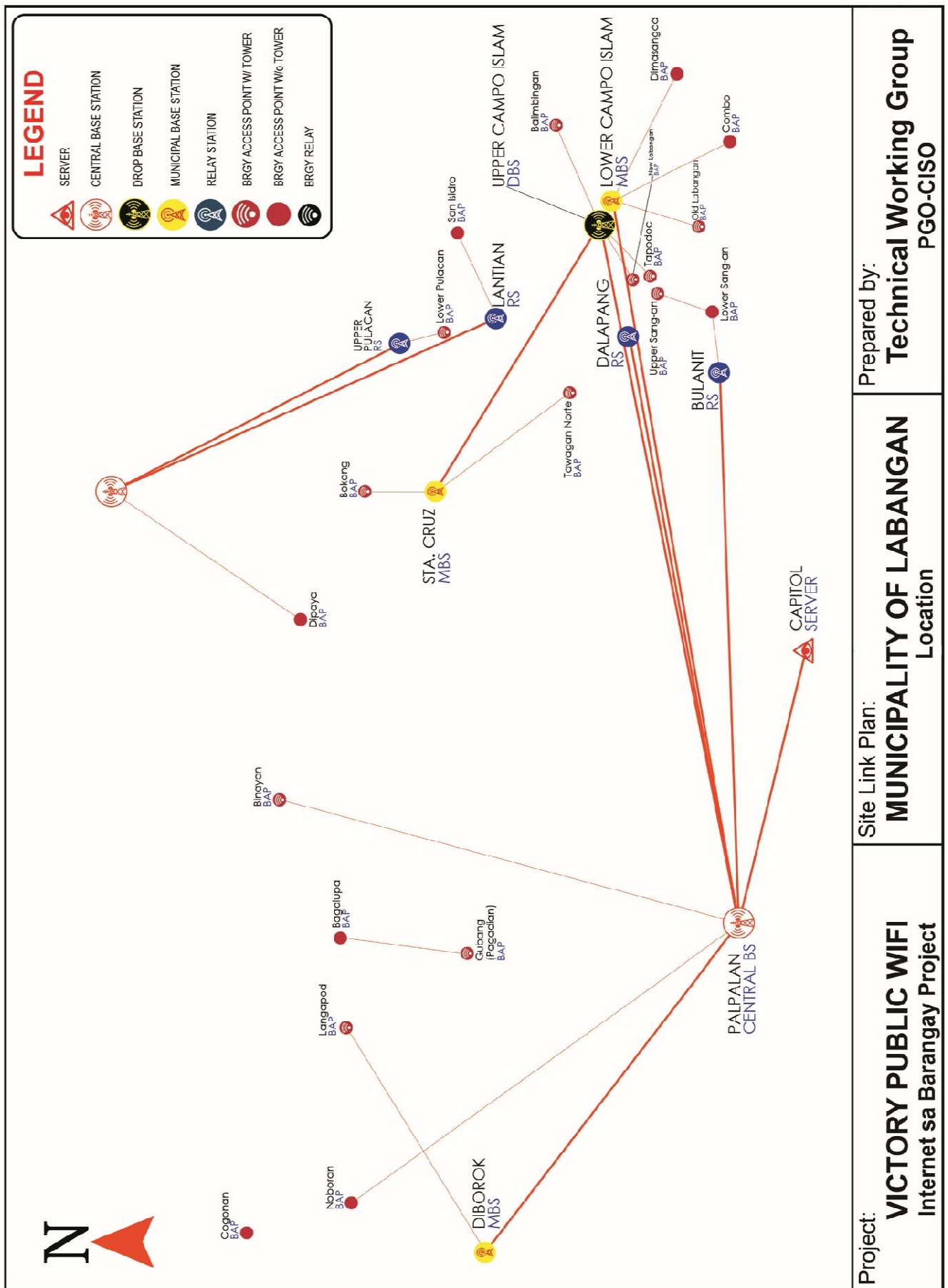
HON. ENGR. VICTOR J. YU

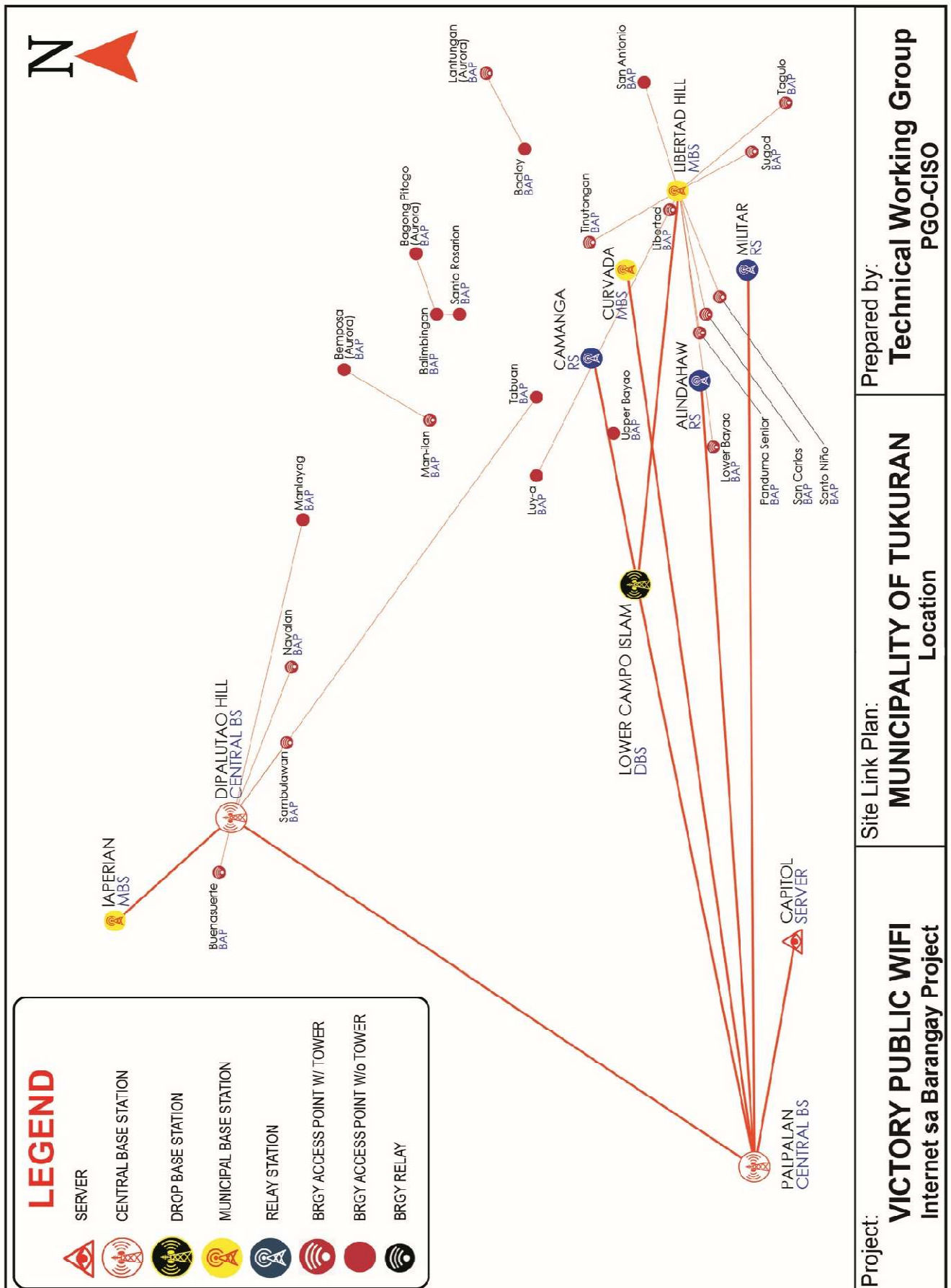
Provincial Governor
Zamboanga del Sur

Attachment 1 - Site Link Plan - Pagadian City

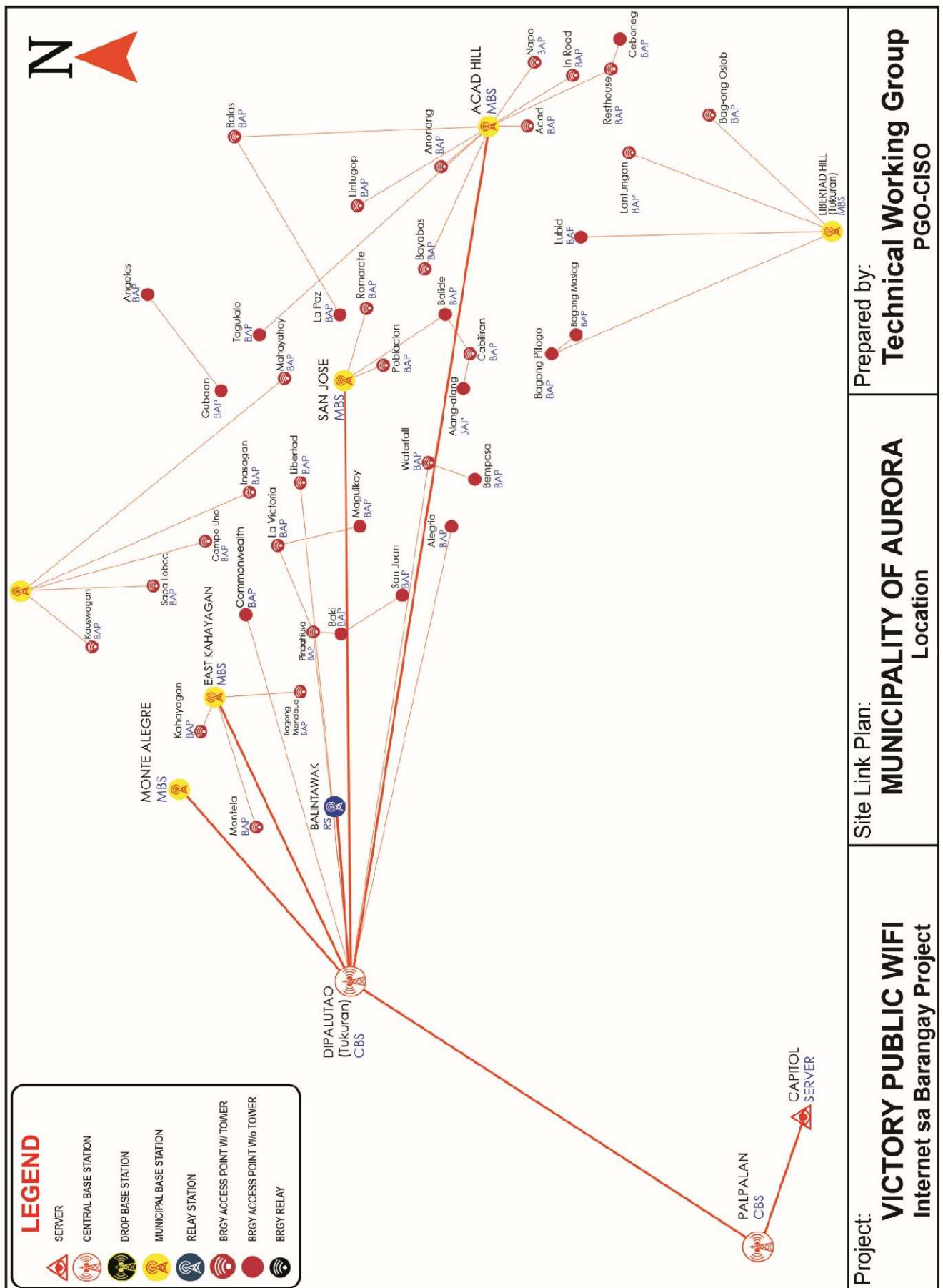


Attachment 2 - Site Link Plan - Labangan

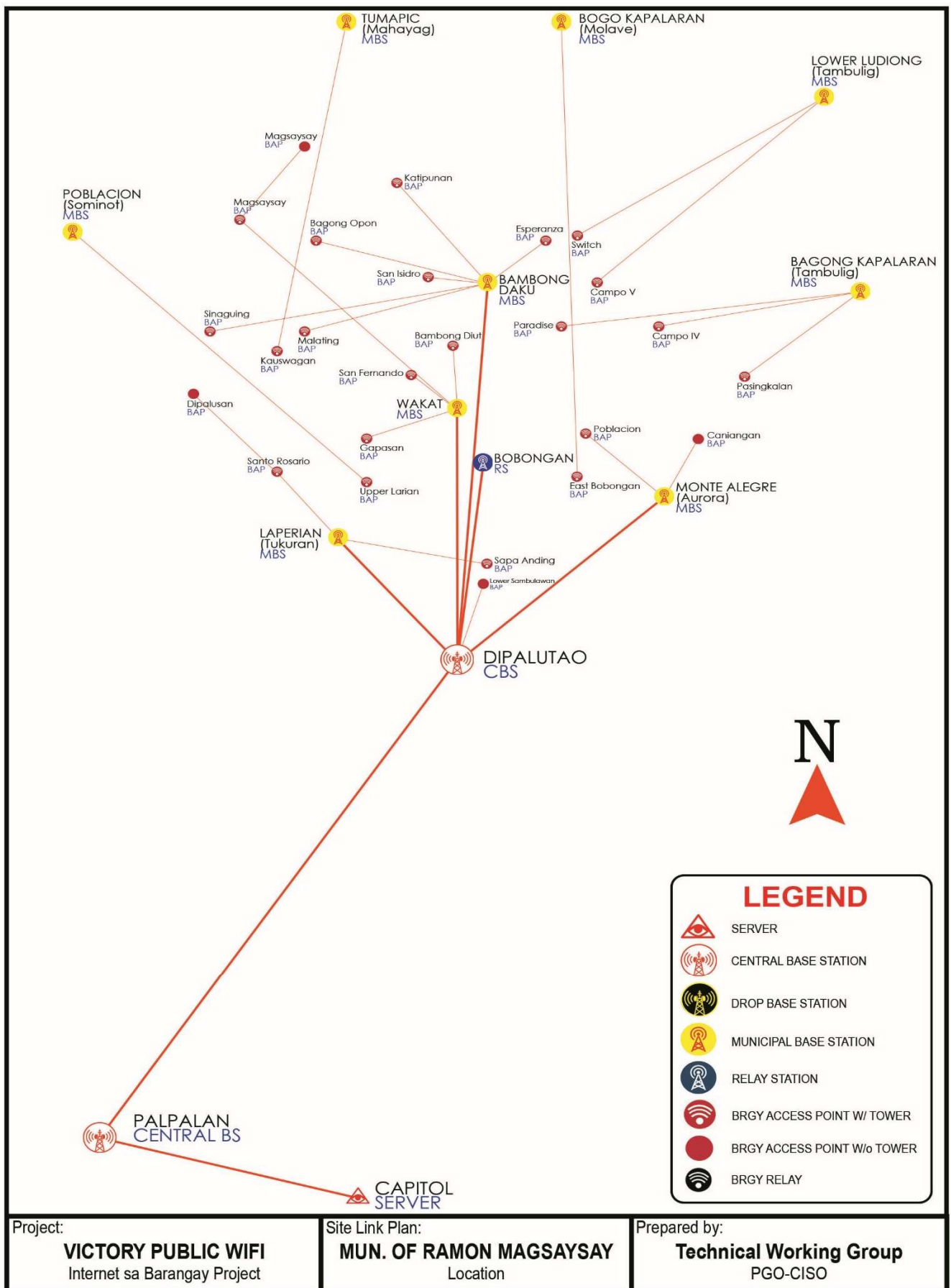




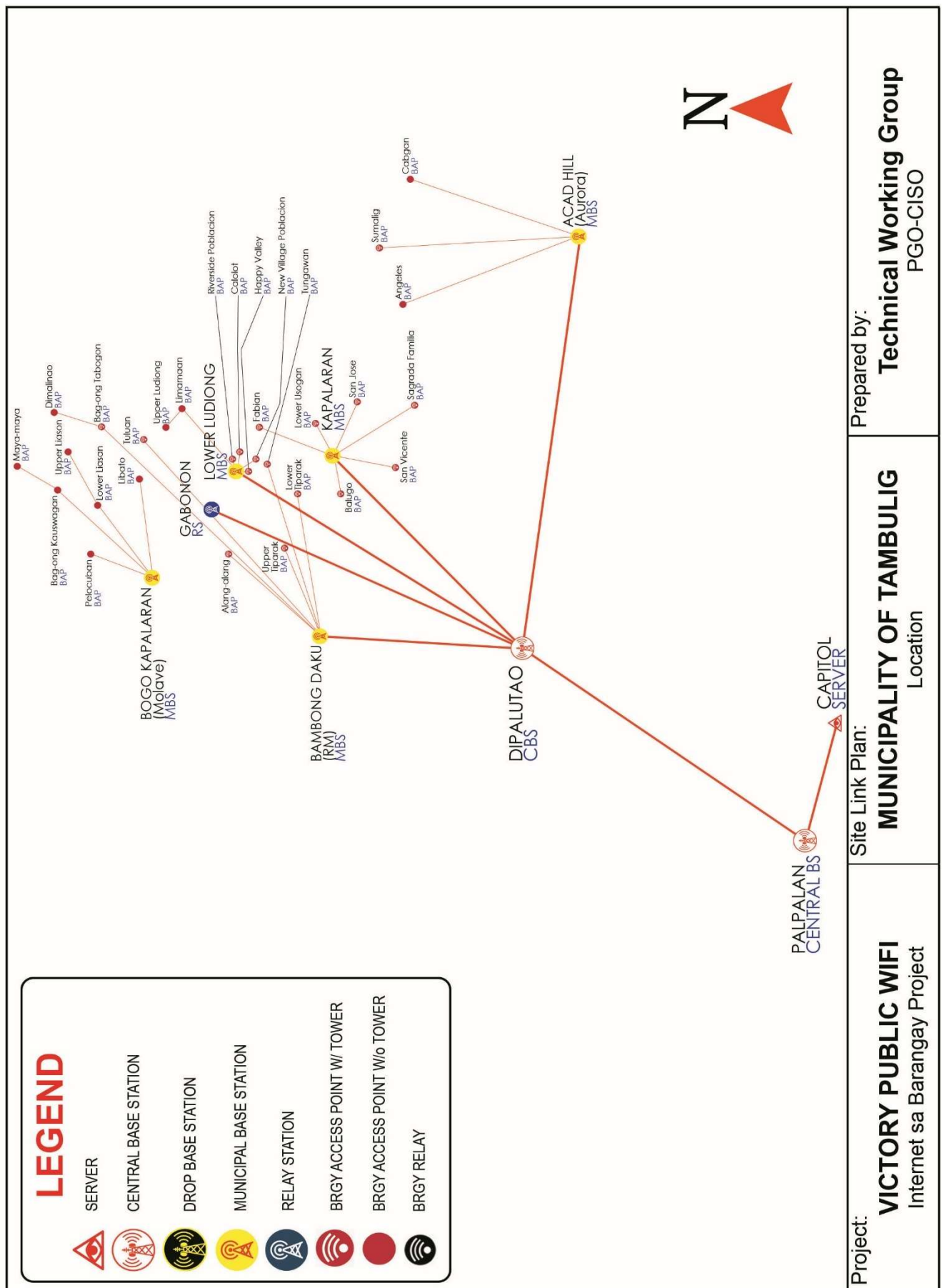
Attachment 4 - Site Link Plan - Aurora



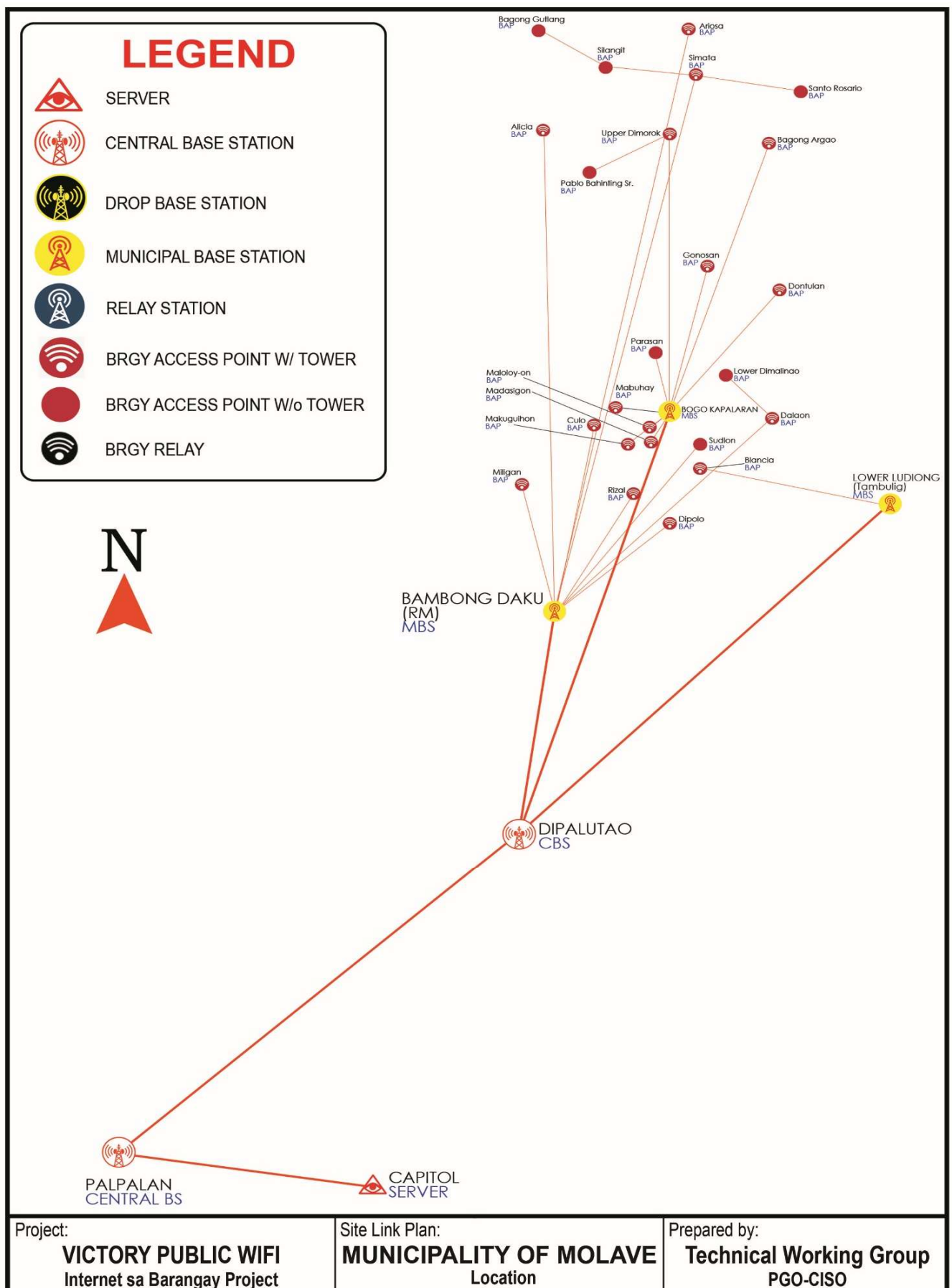
Attachment 5 - Site Link Plan - Ramon Magsaysay



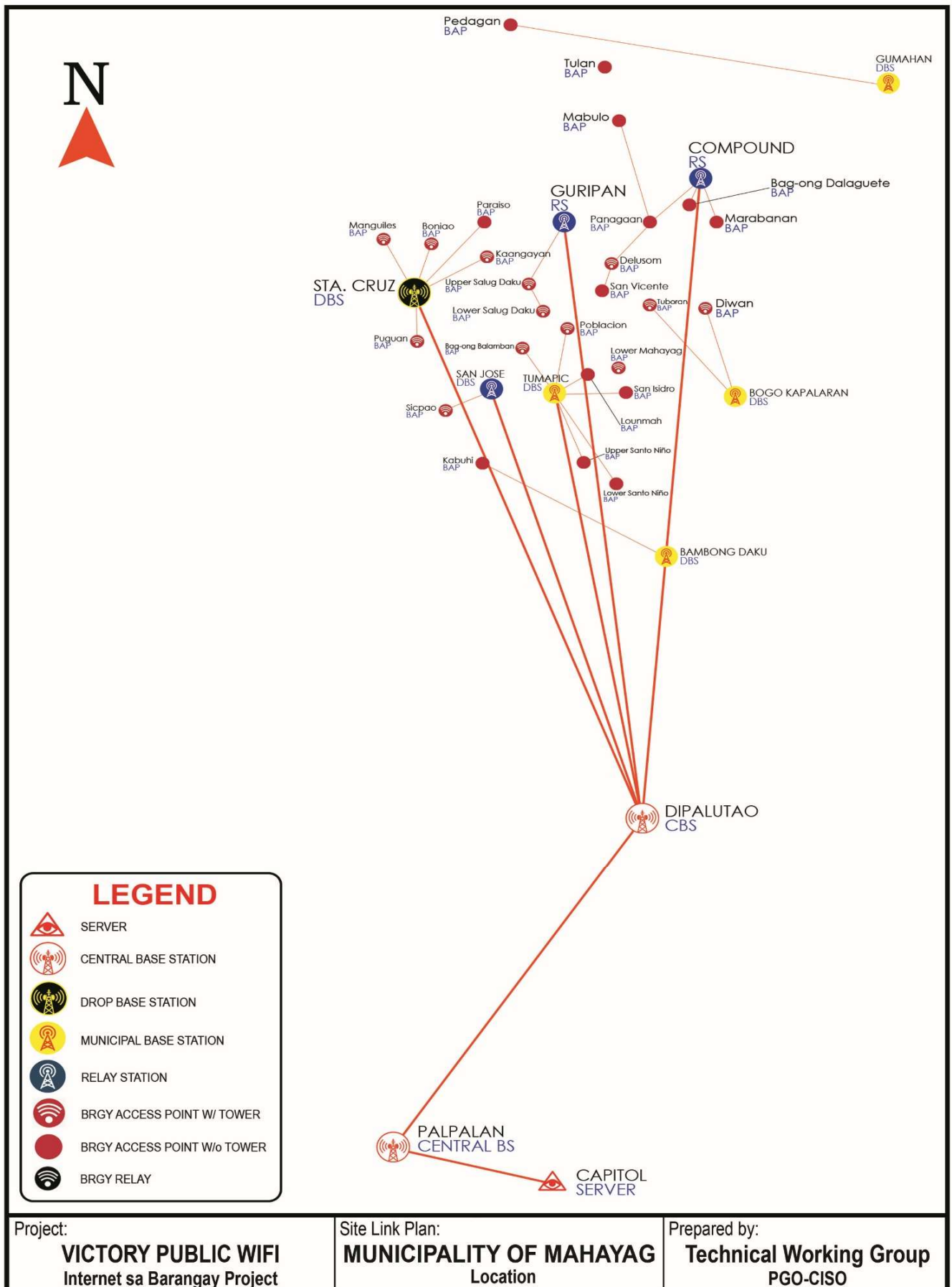
Attachment 6 - Site Link Plan - Tambulig



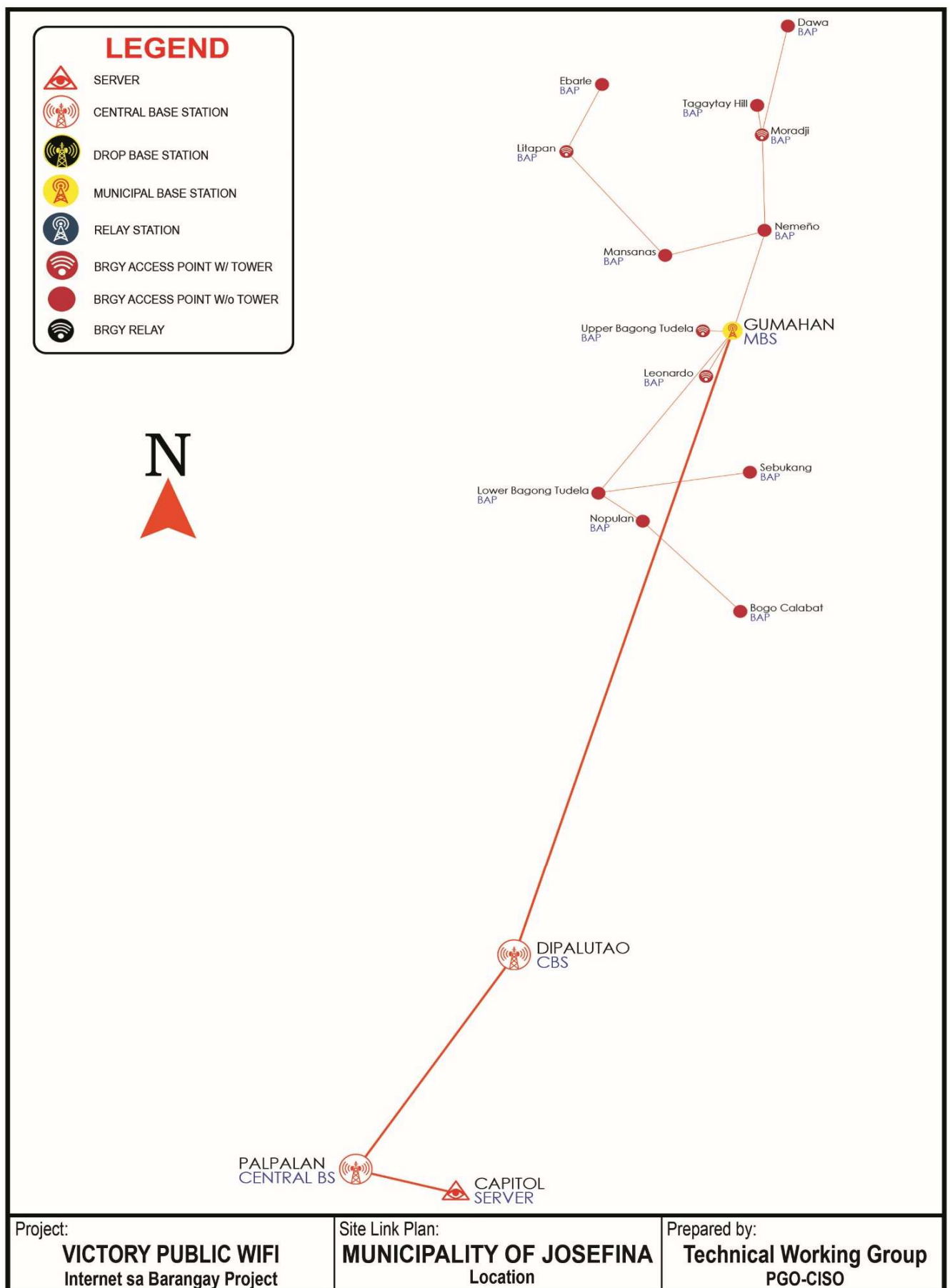
Attachment 7 - Site Link Plan - Molave



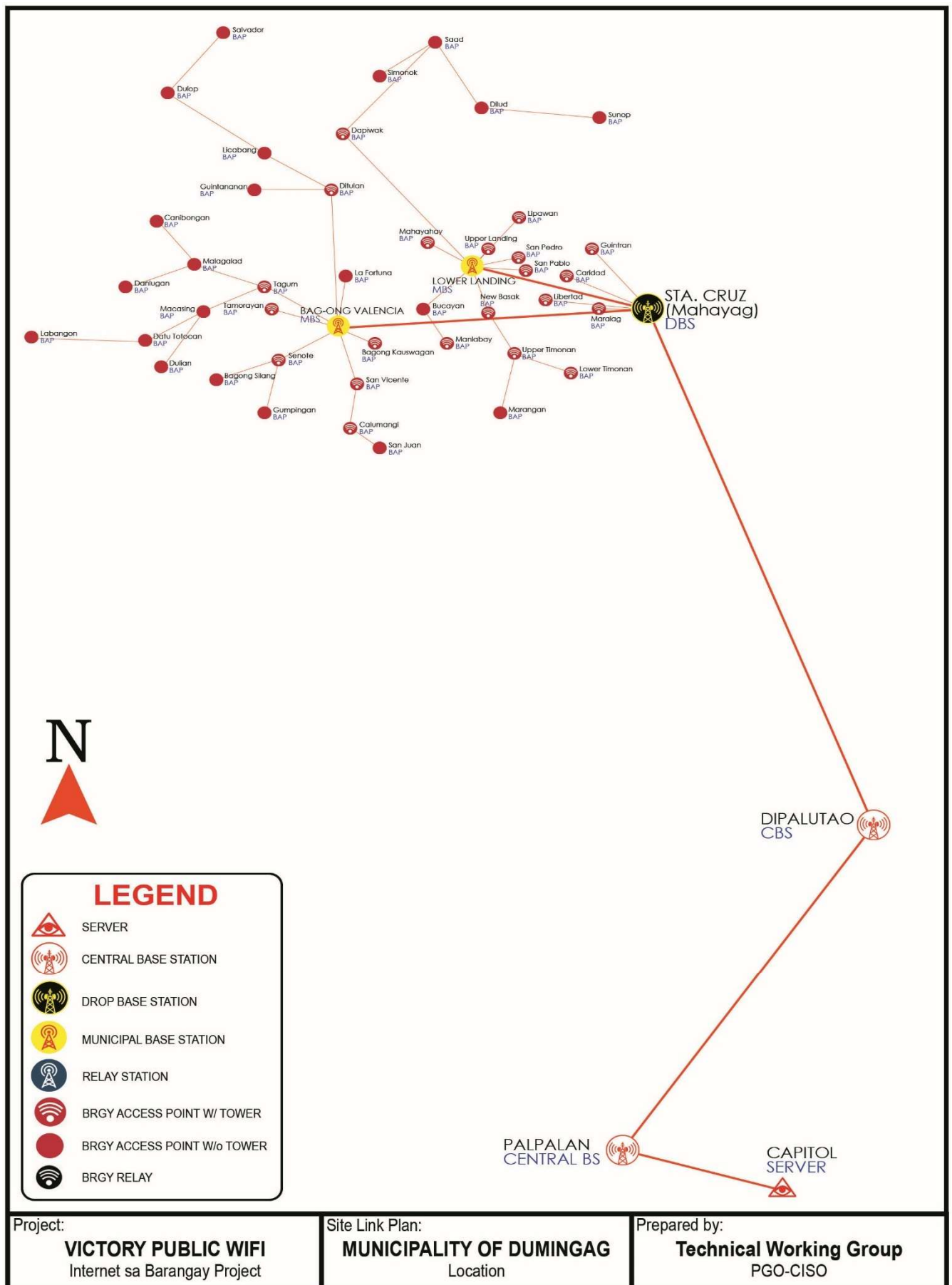
Attachment 8 - Site Link Plan - Mahayag



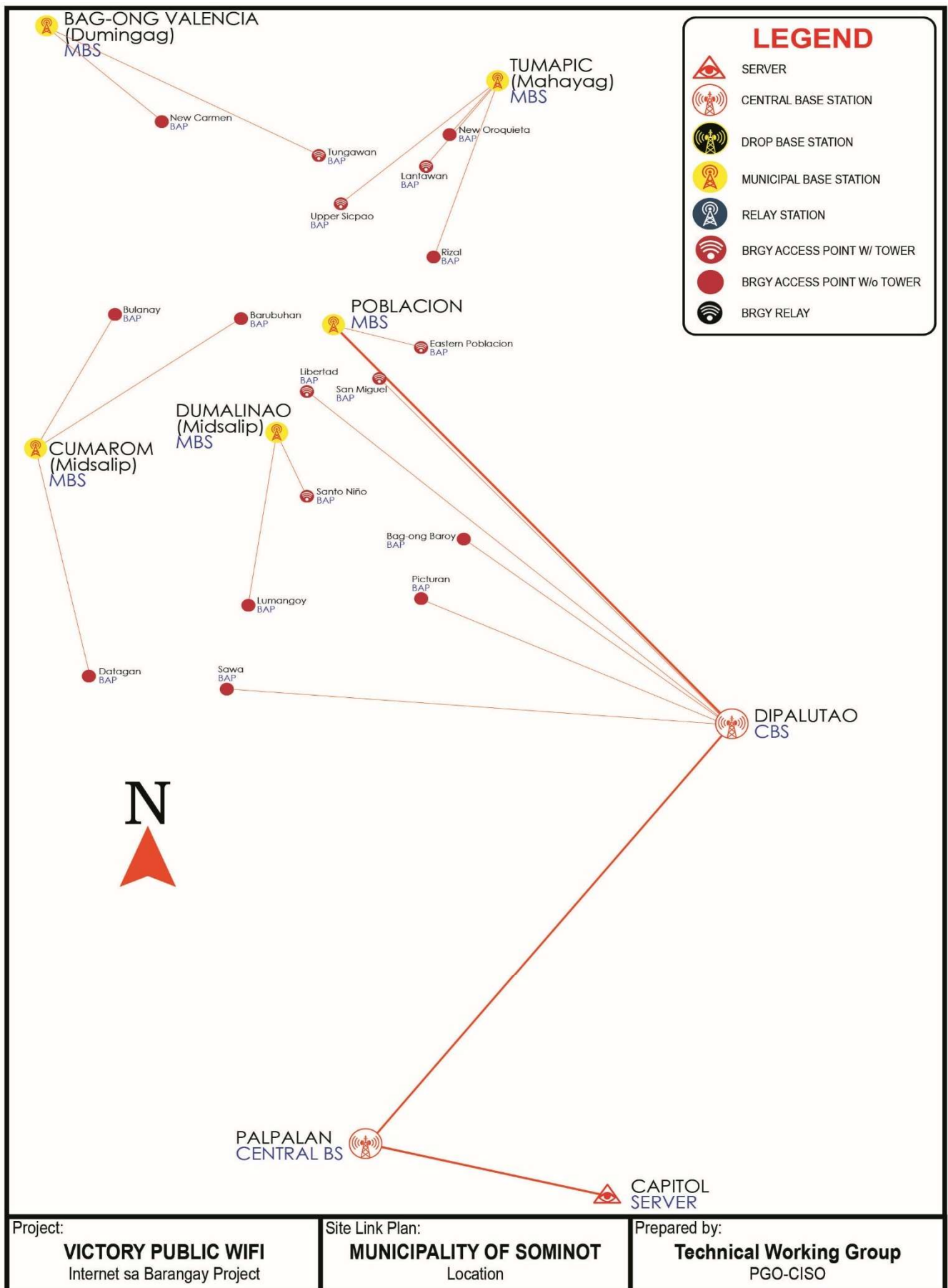
Attachment 9 - Site Link Plan - Josefina



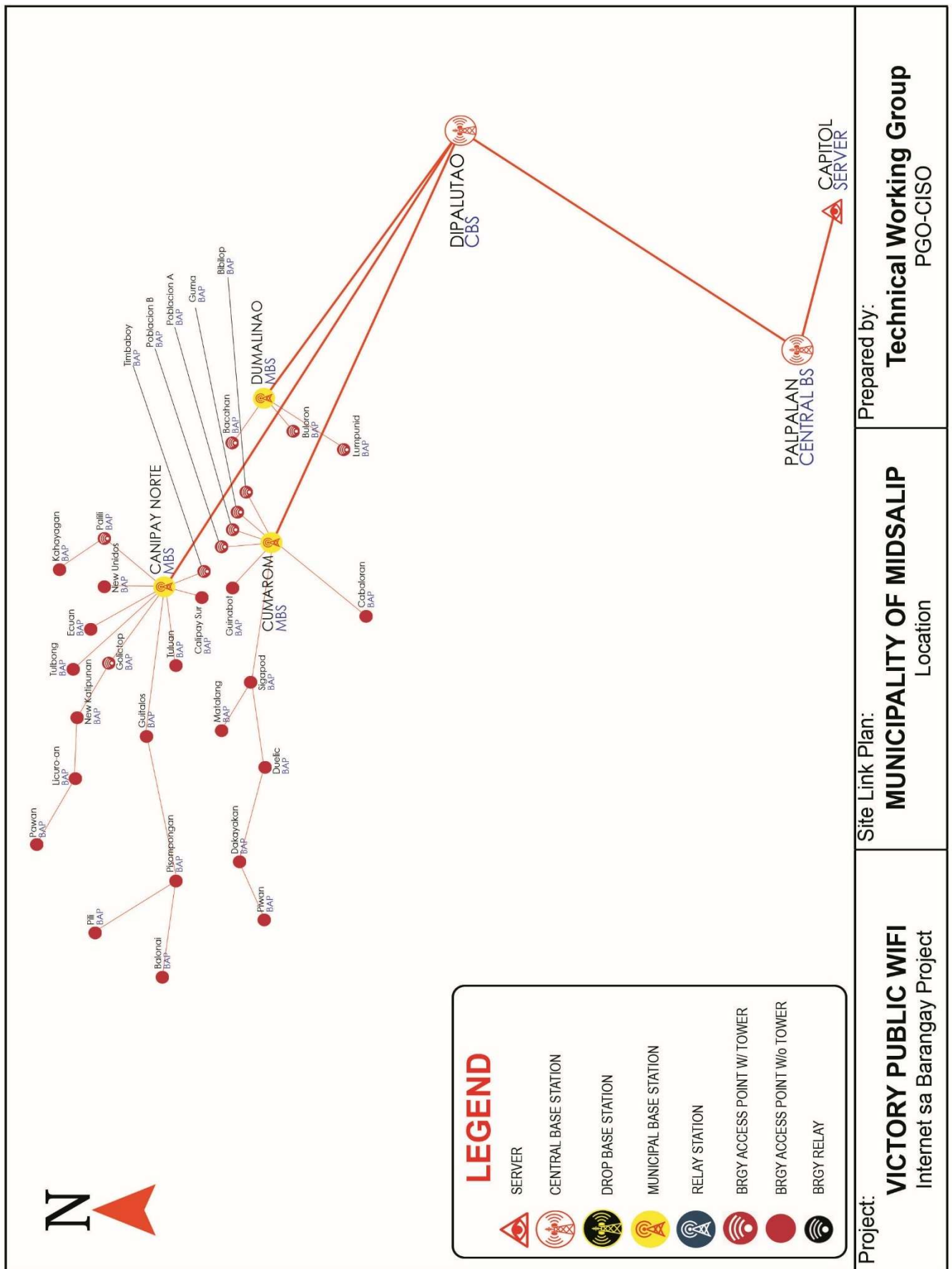
Attachment 10 - Site Link Plan - Dumingag



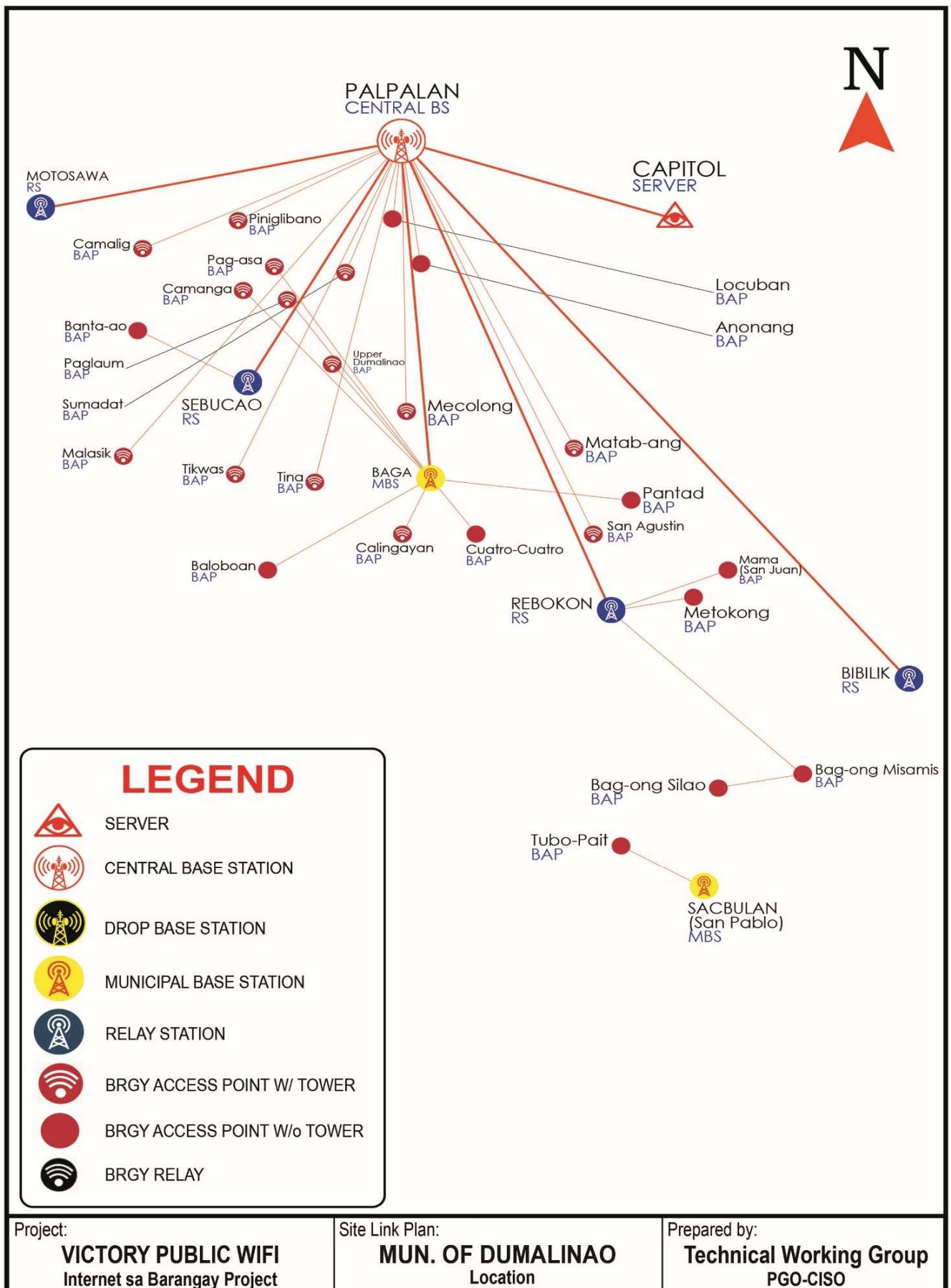
Attachment 11 - Site Link Plan - Sominot



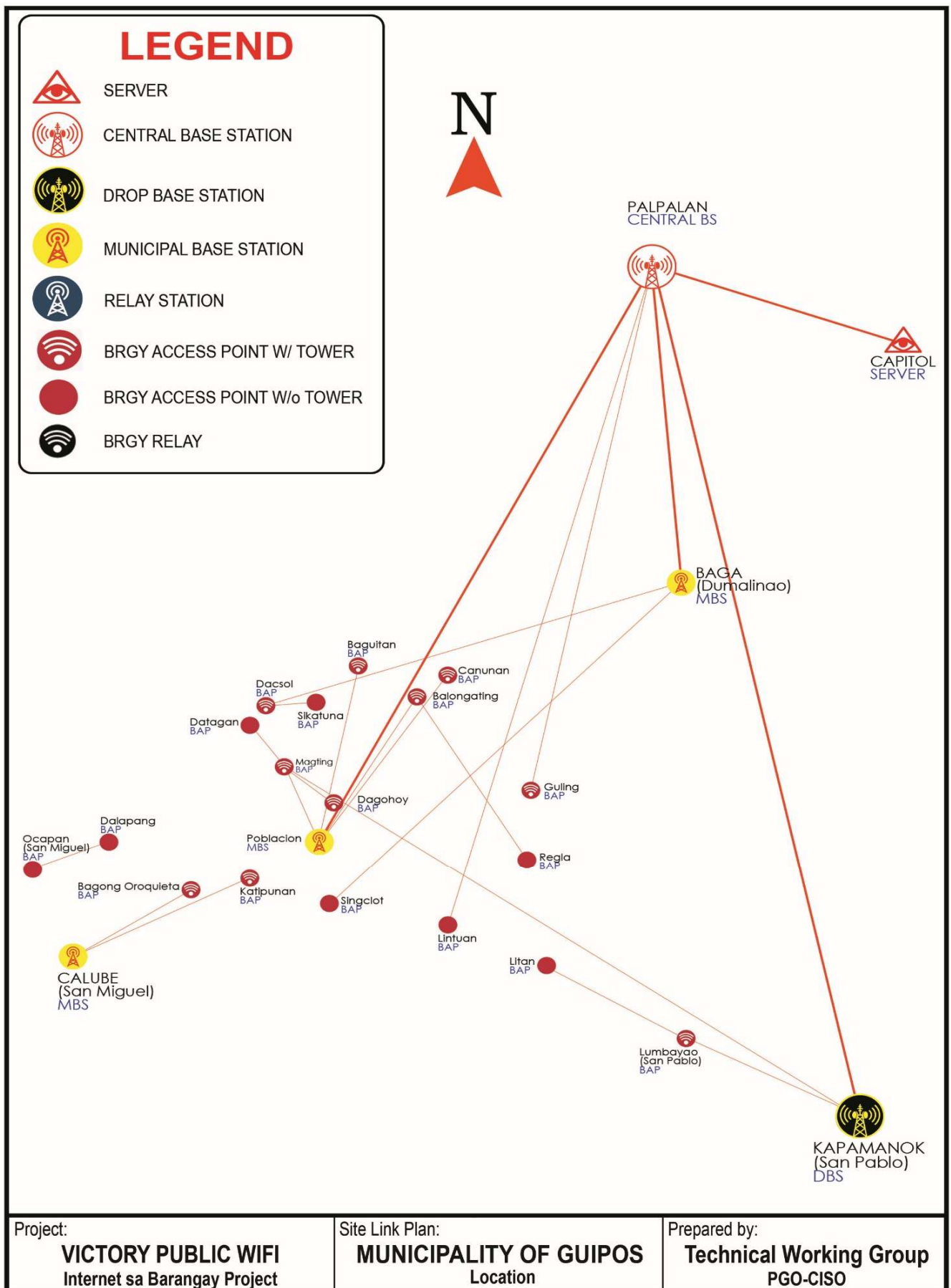
Attachment 12 - Site Link Plan - Midsalip



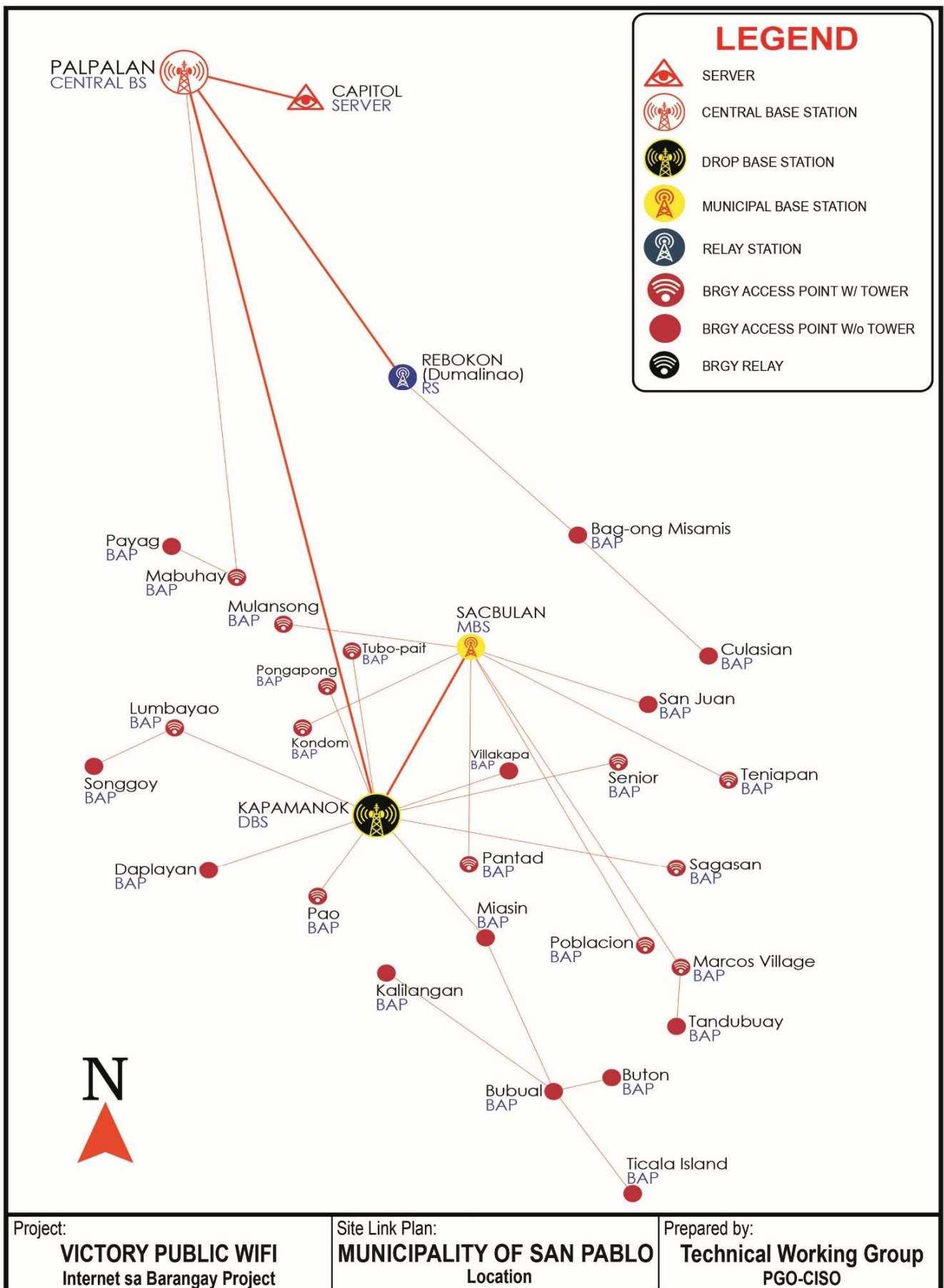
Attachment 13 - Site Link Plan - Dumalinao



Attachment 14 - Site Link Plan - Guipos



Attachment 15 - Site Link Plan - San Pablo



LEGEND

- SERVER
- CENTRAL BASE STATION
- DROP BASE STATION
- MUNICIPAL BASE STATION
- RELAY STATION
- BRGY ACCESS POINT W/ TOWER
- BRGY ACCESS POINT W/o TOWER
- BRGY RELAY

PALPALAN
CENTRAL BASE STATION

PROVINCIAL CAPITOL SERVER

Sagacad Relay Station

Betinan (San Miguel) MBS

Biray BAP

Kinakap BAP

Upper Sibul BAP

Kalian (Margosatubig) MBS

Baluno (Dimataling) MBS

Legarda Tres BAP

Legarda Dos BAP

Legarda Uno BAP

Upper Dimaya BAP

Bacawan BAP

Sumpotan BAP

San Isidro BAP

Ludiong BAP

Ignacio Garrata BAP

Nangka MBS

Lower Dimaya BAP

Proper Dimaya Relay Station

Sambulawan BAP

Locuban BAP

Dongos BAP

Old Mirapao BAP

West Migpulao BAP

East Migpulao BAP

Sungayan BAP

Pisa-an BAP

Guinicolalay BAP

Don Jose BAP

Poblacion BAP

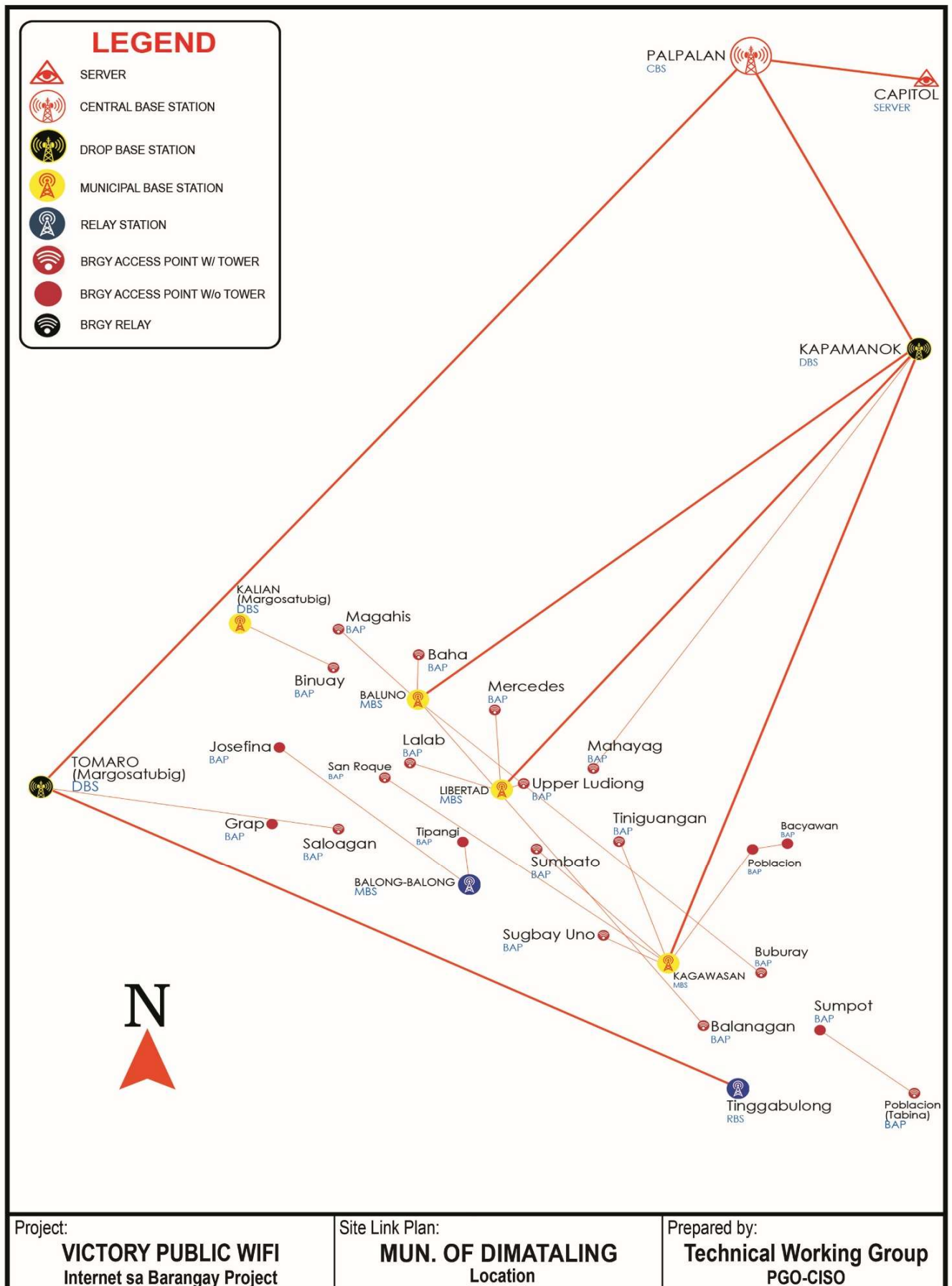
Benuatan BAP

Nian BAP

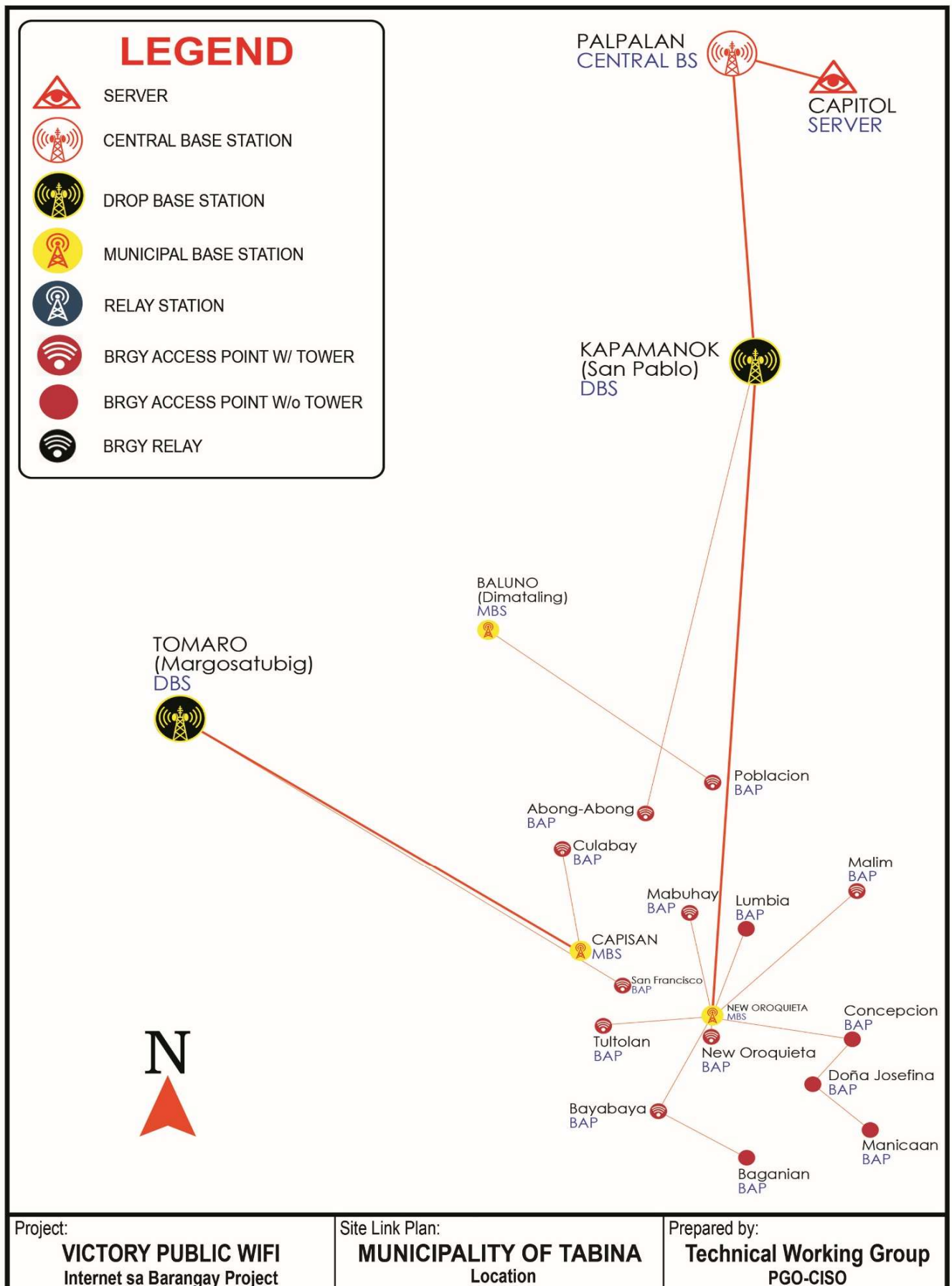
Tarakan BAP

Prepared by:
Technical Working Group
PGO-CISO

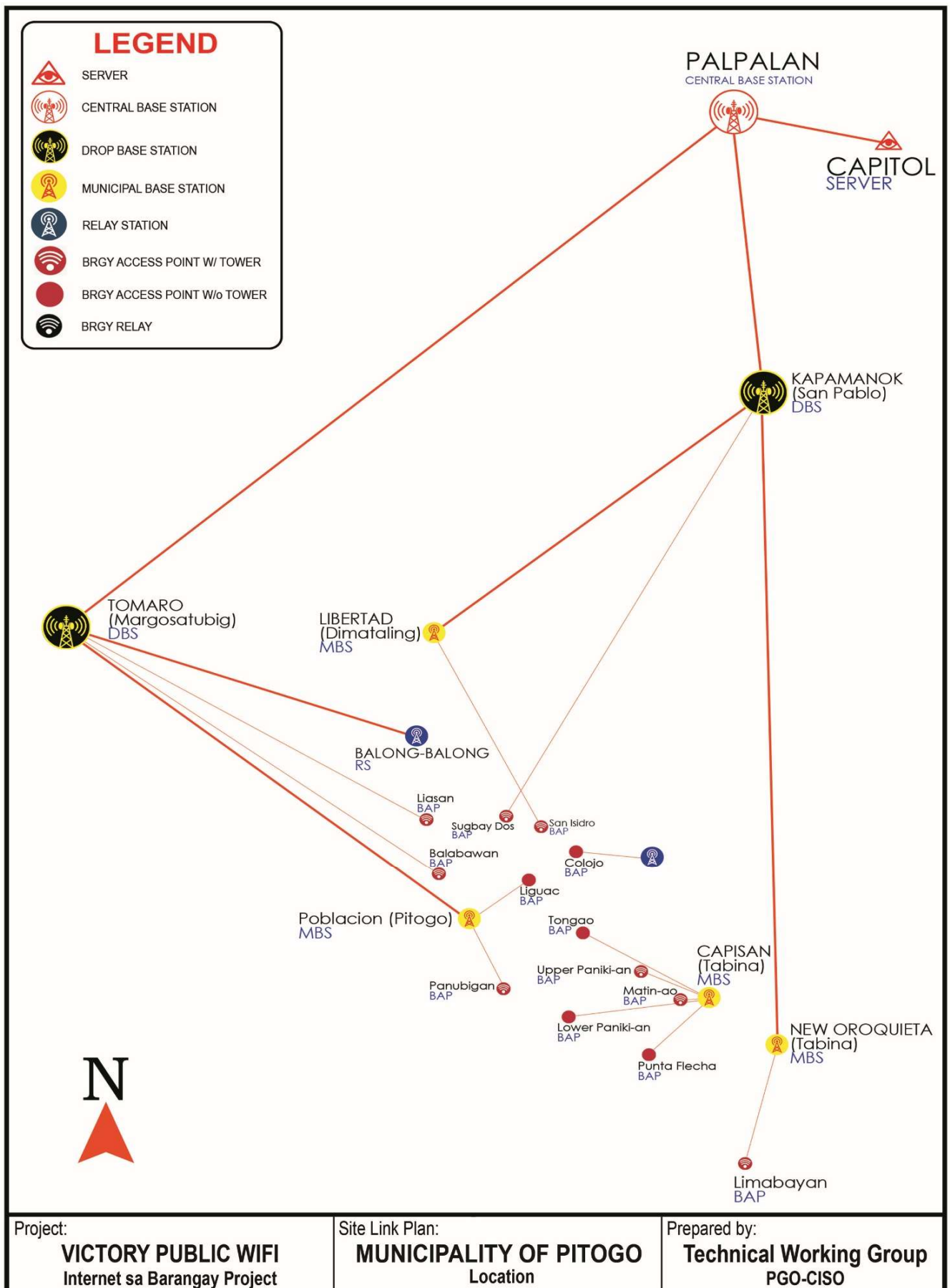
Attachment 17 - Site Link Plan - Dimataling



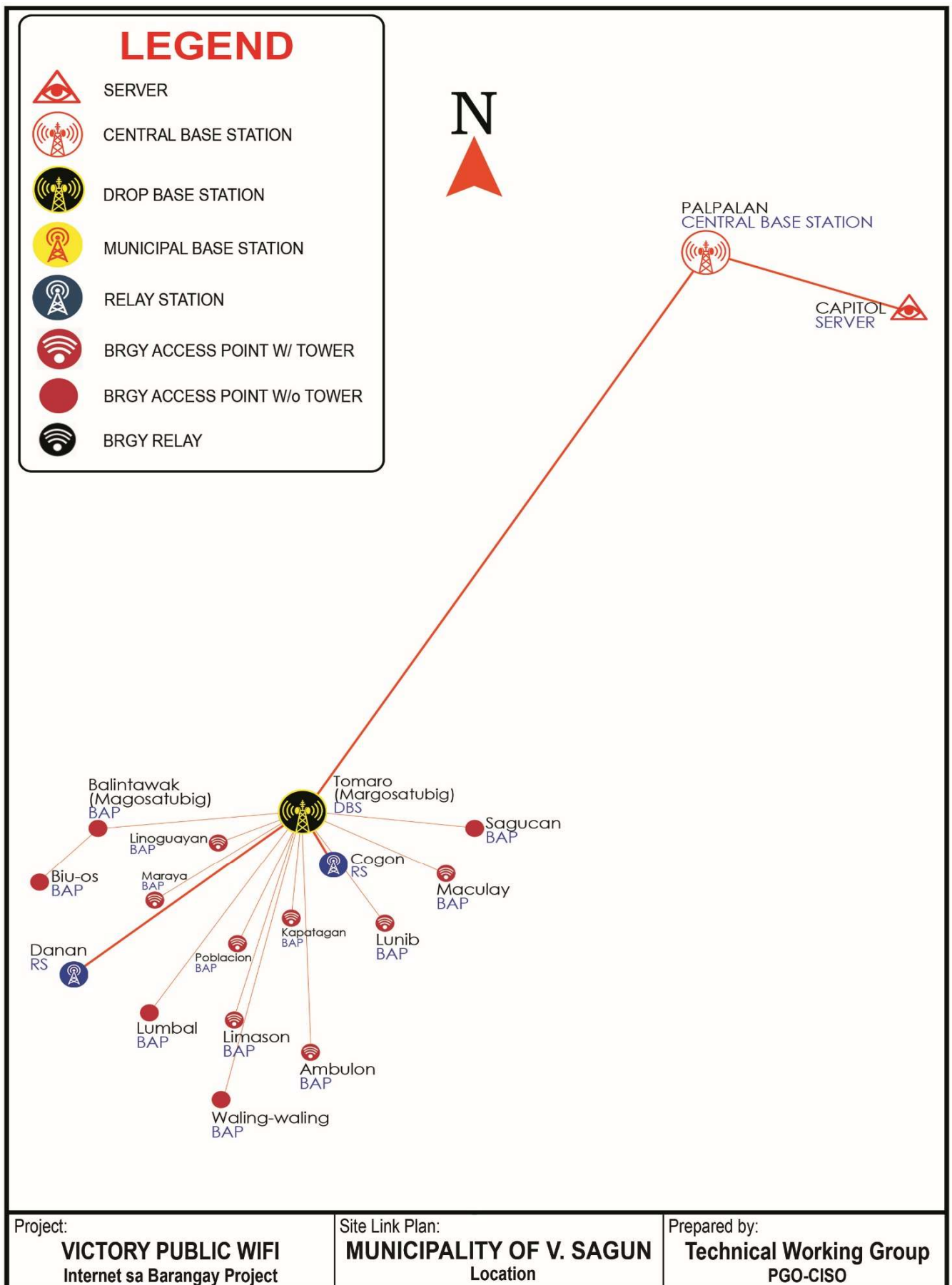
Attachment 18 - Site Link Plan - Tabina



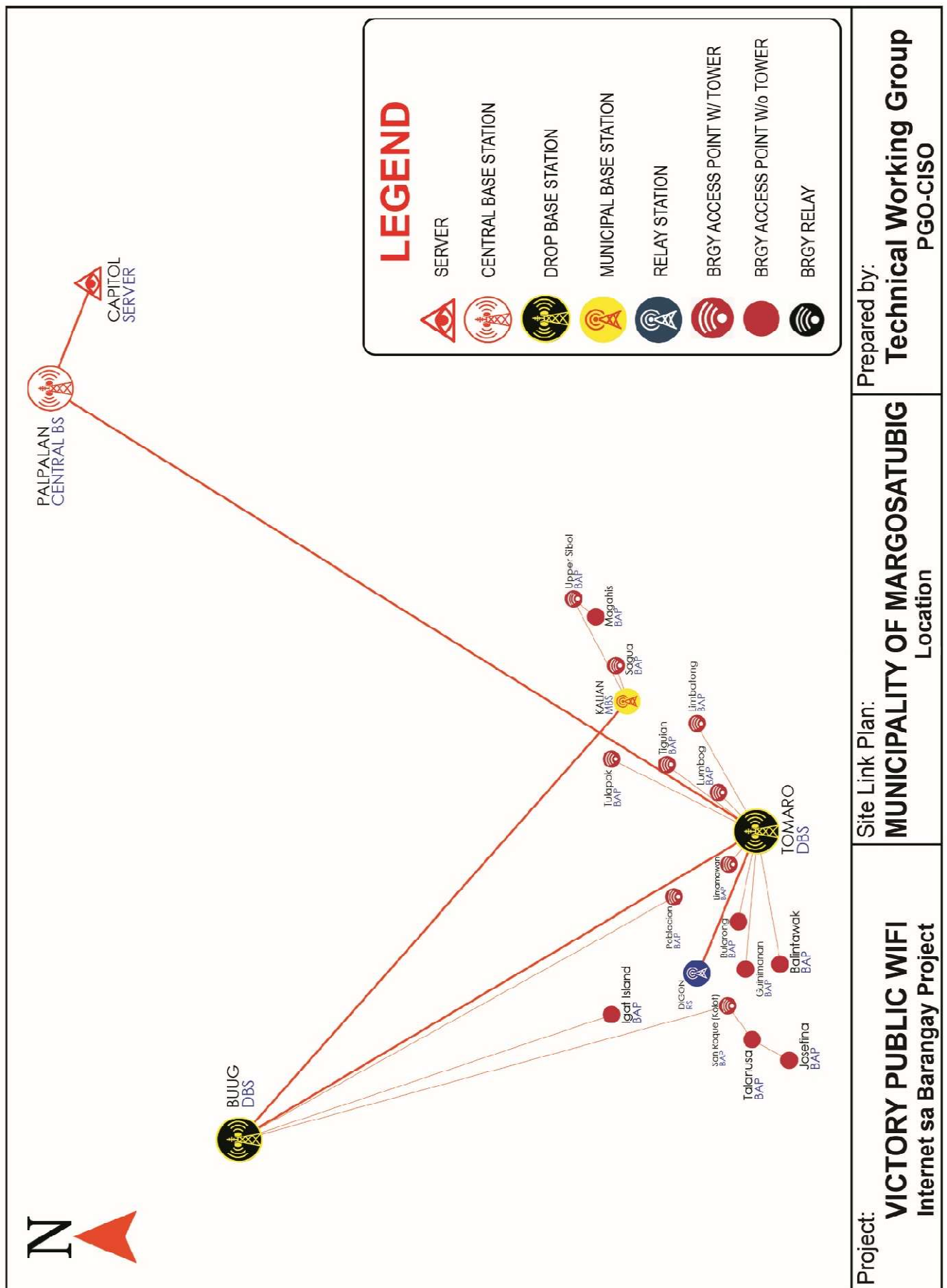
Attachment 19 - Site Link Plan - Pitogo



Attachment 20 - Site Link Plan - Vincenzo Sagun



Attachment 21 - Site Link Plan - Margosatubig



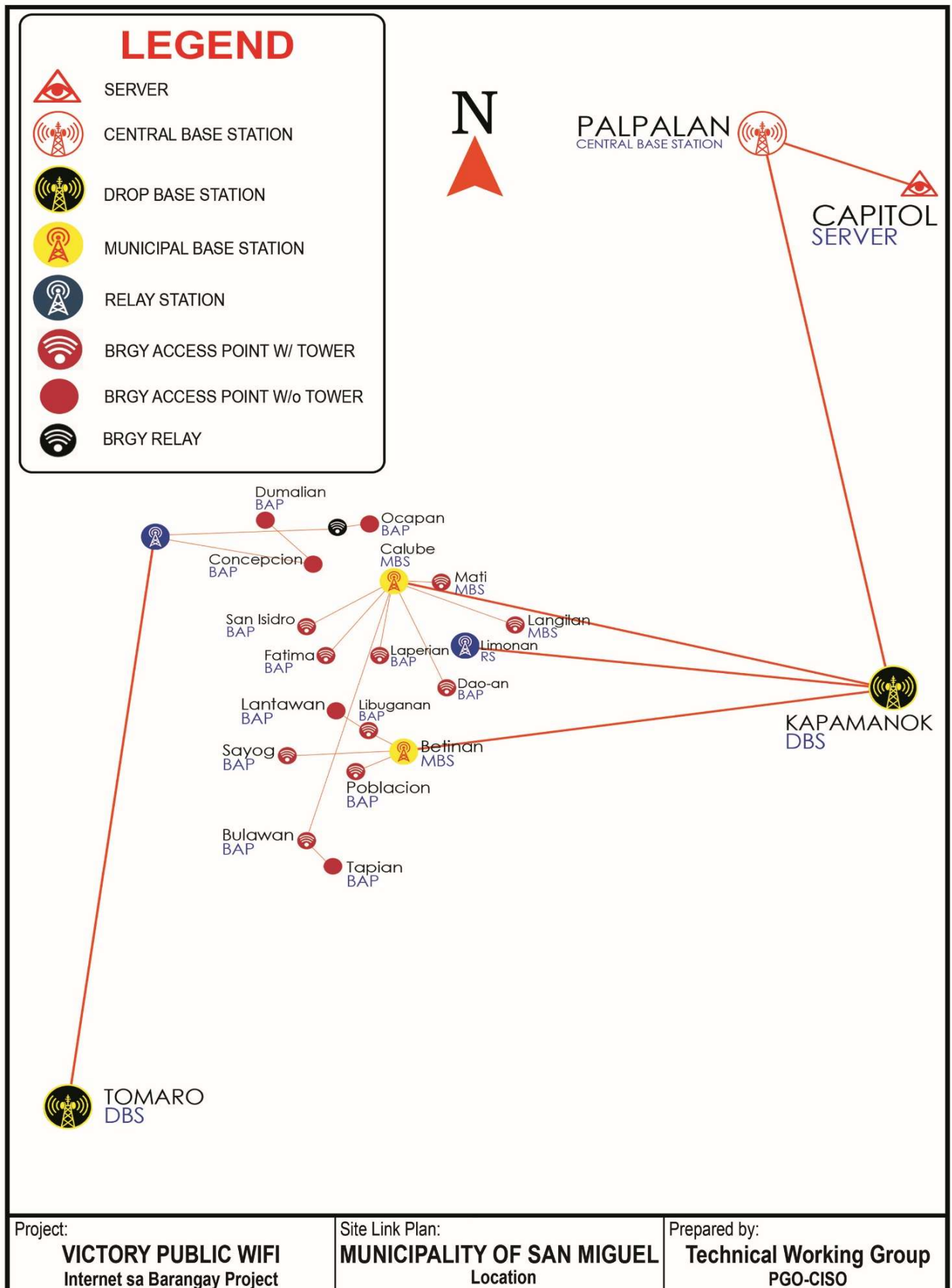
LEGEND

- SERVER
- CENTRAL BASE STATION
- DROP BASE STATION
- MUNICIPAL BASE STATION
- RELAY STATION
- BRGY ACCESS POINT W/ TOWER
- BRGY ACCESS POINT W/o TOWER
- BRGY RELAY

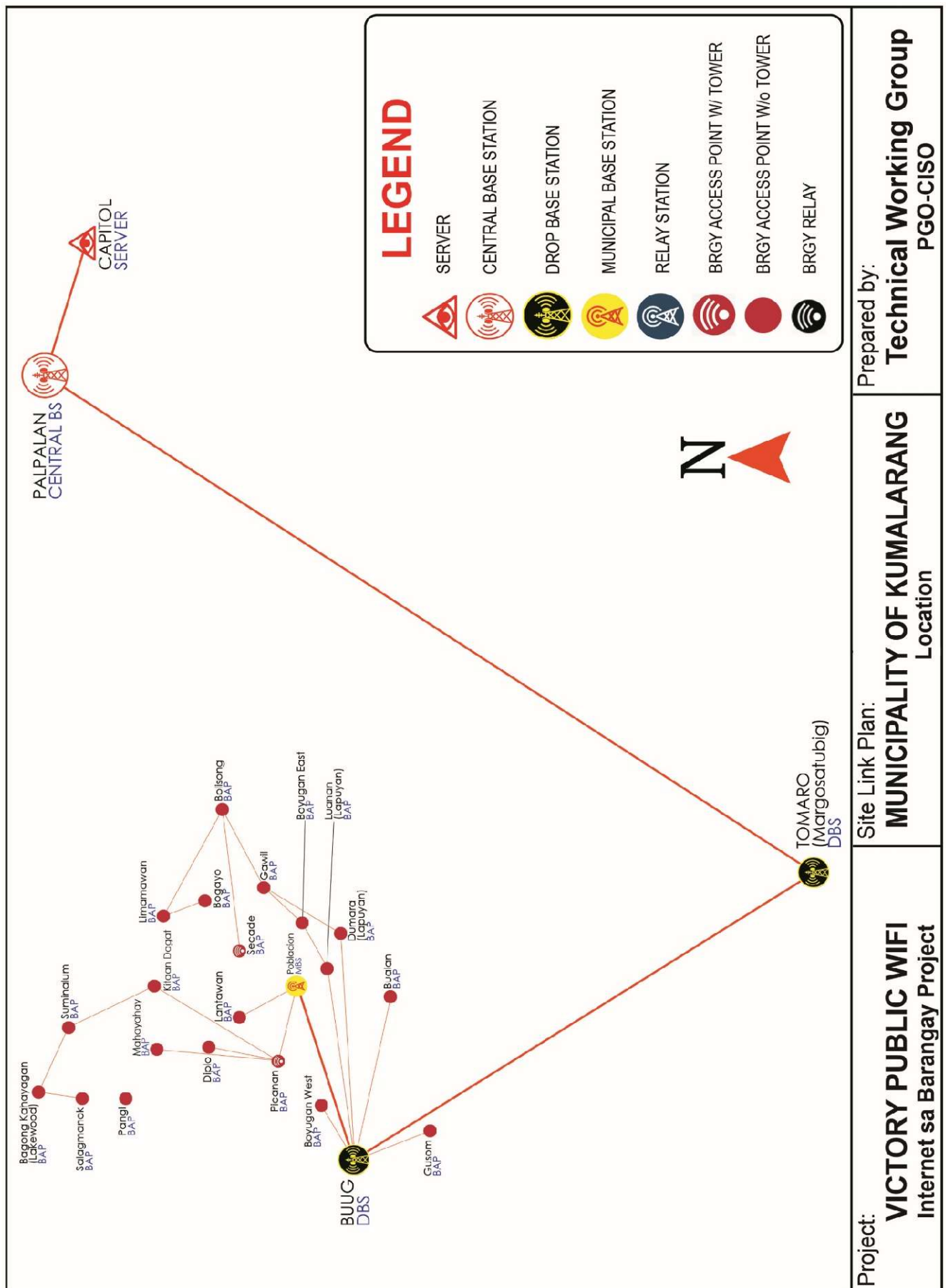
Locations:

- BUUG DBS
- PAIPALANI CENTRAL BS
- PROVINCIAL CAPITOL SERVER
- Concepcion (San Miguel)
- Marilag BAP
- Taigabob BAP
- Tiguhia RS
- San Jose BAP
- Dumara BAP
- Lucnan BAP
- Mandeg BAP
- Tabon BAP
- Pingalay BAP
- Mahalingeb BAP
- Karpok BAP
- Pantao BAP
- TINANGALANG MS
- Lenok BAP
- Danganan BAP
- Pampang BAP
- Poblacion BAP
- Salamatayan RS
- Tiguan BAP
- Dansal BAP
- Sayag BAP
- Lubusan BAP
- Bulawan BAP
- Molum BAP
- KALIAN (Margosatubig) MS
- TOMARO (Margosatubig) DBS
- Igat Island (Margosatubig) BAP

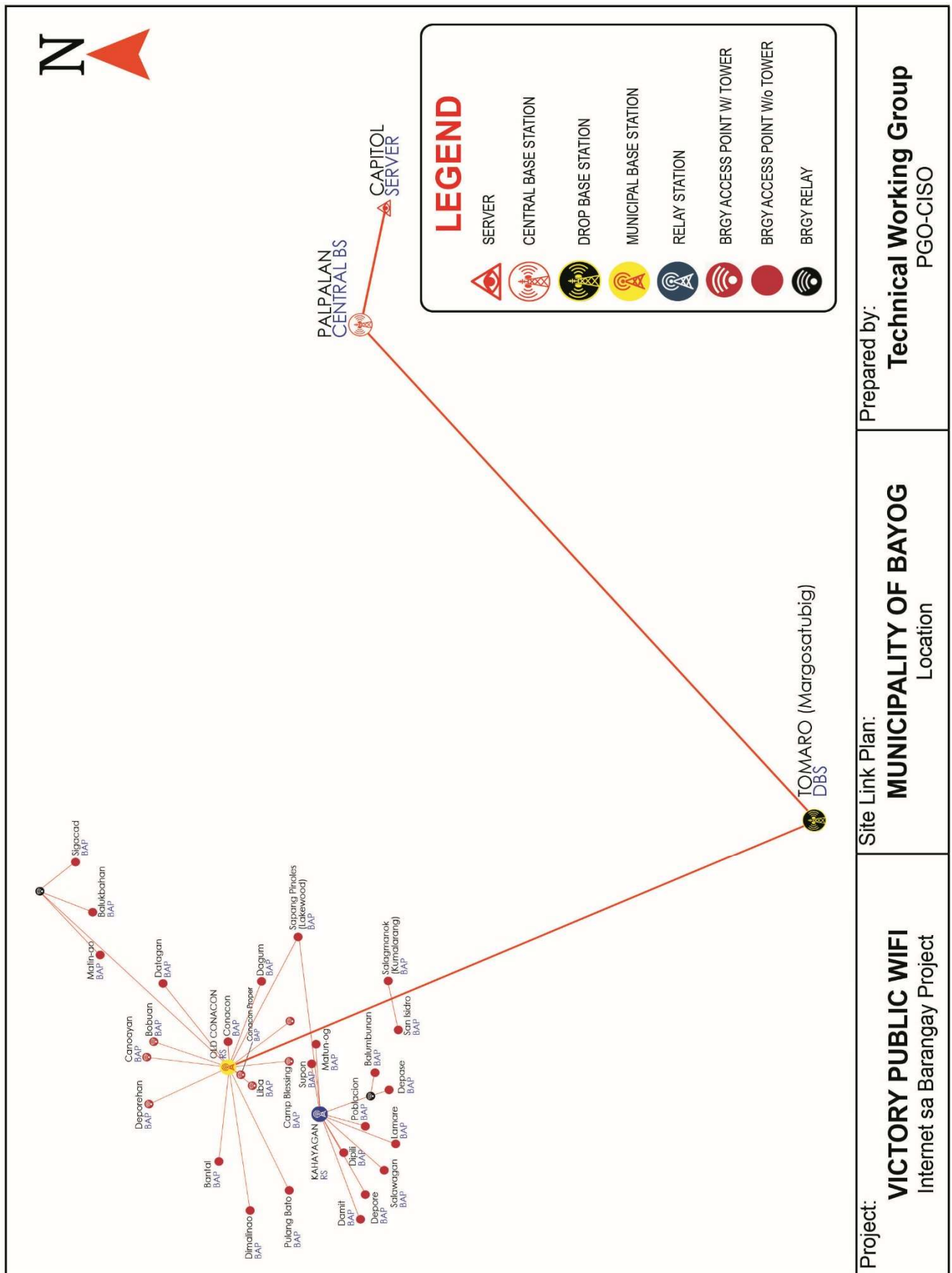
Attachment 23 - Site Link Plan - San Miguel



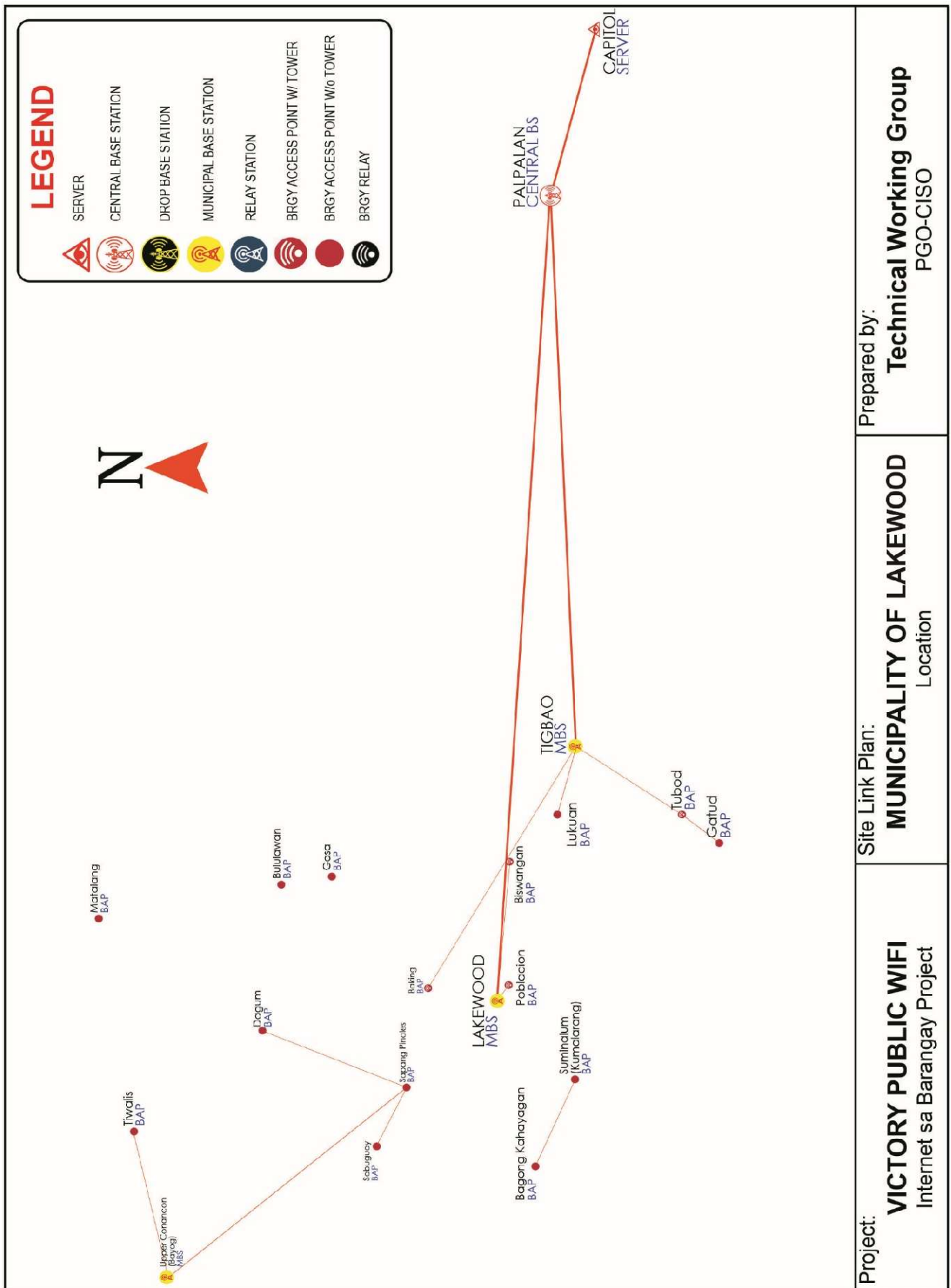
Attachment 24 - Site Link Plan - Kumalarang



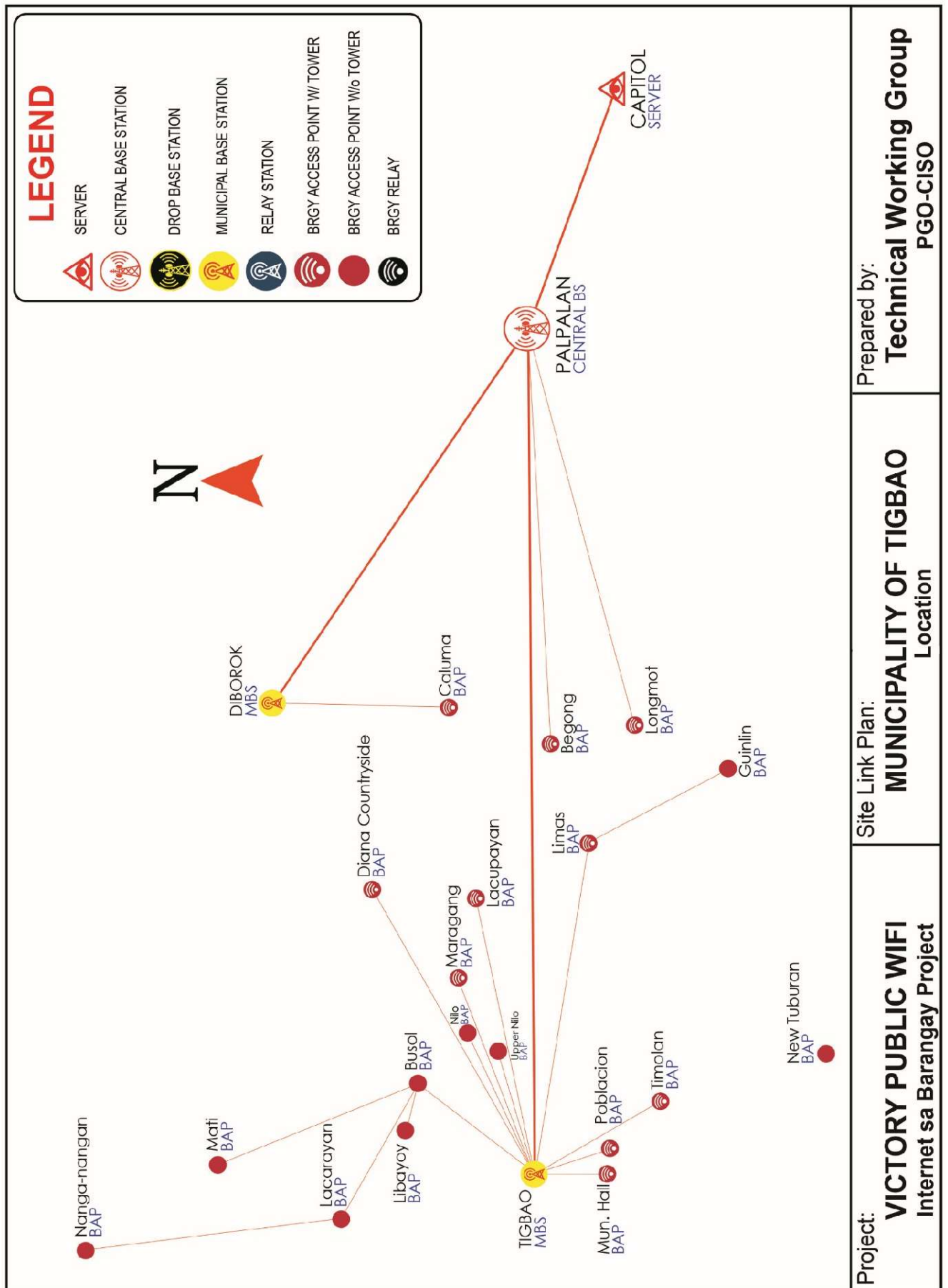
Attachment 25 - Site Link Plan - Bayog



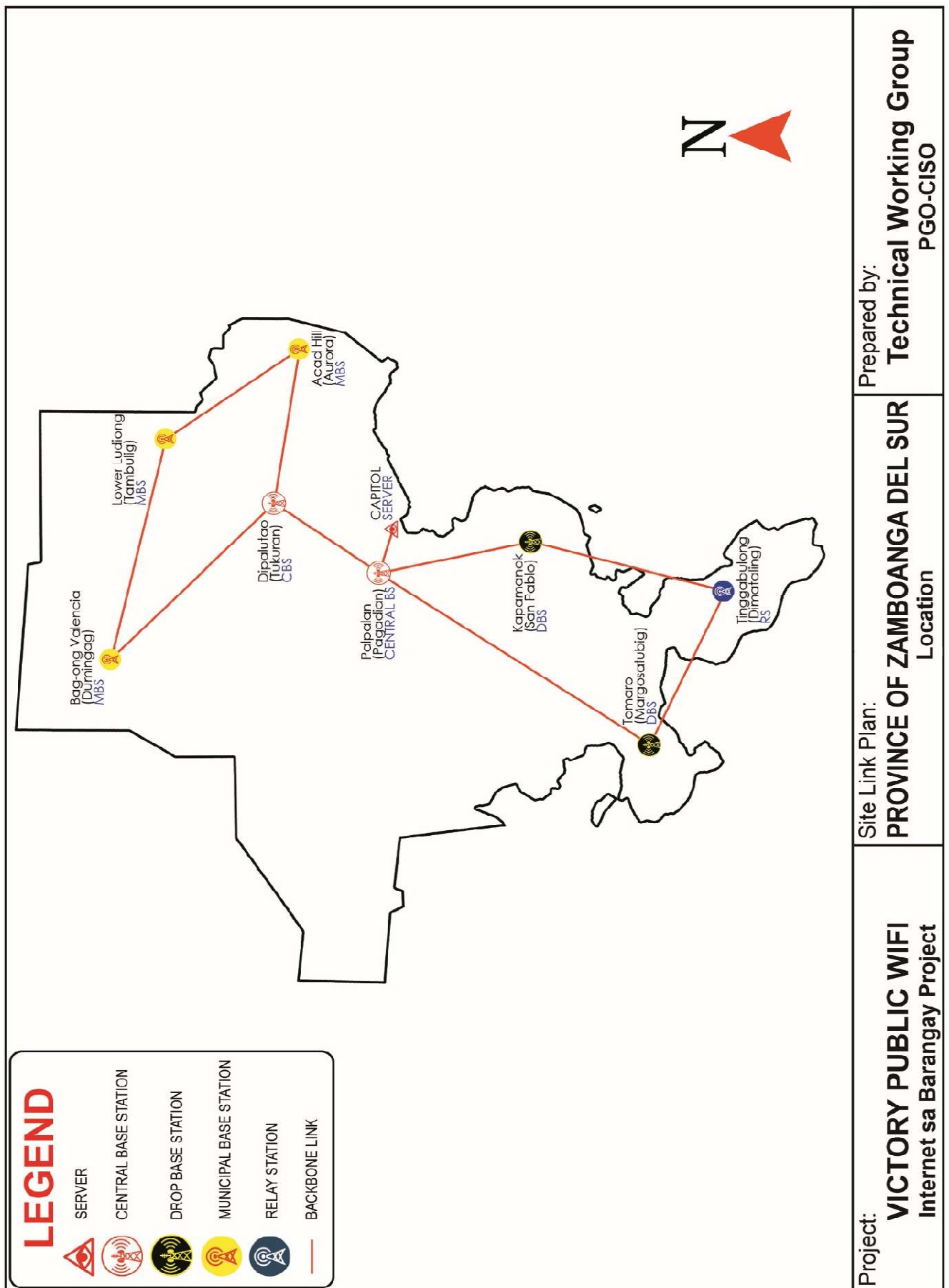
Attachment 26 - Site Link Plan - Lakewood



Attachment 27 - Site Link Plan - Tigbao



Attachment 28 - Site Link Plan - Backbone Link



Attachment 29 - **SAMPLE CAPTIVE PORTAL**

VICTORY FREEWiFi
ACCESS SERVICE

PUBLIC WIFI

Please enter your details below

Email Address

First Name

Last Name

Please select your age group ▼

CONTINUE

☐ Please confirm you agree to the Terms and Conditions.

☐ Yes, I would like to receive news, offers and promotions

[View our Terms & Conditions](#)

VICTORY FREEWiFi **victoryfreewifi.net**